A Supervised Machine Learning Framework for Risk-Based Audit Selection in the South Sudan Revenue Authority

ANYIETH AGUEK AYUEN

VU-MCS-2407-0721-WEE

+211 927 317 297

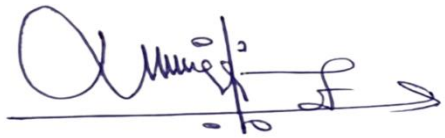
makueireng98@gmail.com

"This dissertation is submitted in partial fulfilment of the requirements for the award of a Master of Science in Computer Science at Victoria University, Uganda."

DECLARATION

I, Anyieth Aguek Ayuen, declare that this dissertation is my original work. It has not been presented and will not be presented to any other University for a similar or any other educational award.

Signature:



Anyieth Aguek Ayuen.

VU-MCS-2407-0721-WEE

APPROVAL

This dissertation is submitted to the Faculty of Science and Technology of Victoria University Uganda with my approval as the research supervisor.

Signature:



Name:

Mr. Bua Anthony

DEDICATION

This dissertation is affectionately dedicated to my beloved parents, who made countless sacrifices to lay the foundation of my education. Your prayers, resilience, and lifelong belief in the value of knowledge have been my guiding light.

To my wonderful wife, for your endless patience, unconditional love, and constant encouragement. You held our world together during the late nights and long hours of research, and this achievement belongs to you as much as it does to me.

To the management of the South Sudan Revenue Authority (SSRA), for their foresight and trust in granting me permission to utilize their institutional data for this study.

And to Victoria University, for providing the academic sanctuary and mentorship that transformed my curiosity into structured, real-world innovation.

ACKNOWLEDGEMENT

I would like to express my profound gratitude to the South Sudan Revenue Authority (SSRA) for providing access to the historical taxpayer datasets, operational ledger metadata, and administrative parameters that were essential in validating this machine learning framework.

Special recognition is extended to the Faculty of Science and Technology and the Department of Computer Science at Victoria University Uganda for their rigorous academic mentorship, technical guidance, and continuous support throughout this research endeavour. I am particularly grateful to my supervisor, Mr. Anthony Bua, for his invaluable mentorship, patience, constructive feedback, and unwavering guidance throughout the entire research process.

I would also like to extend my sincere appreciation to the Commissioner of the Information Technology & Digital Transformation Division, Mr. Garang Maker Thiong, of the South Sudan Revenue Authority, for his steadfast support and for granting the necessary permissions during the process of data acquisition, model training, and validation. His encouragement and confidence in this work were instrumental to its successful completion.

My heartfelt gratitude goes to my beloved parents, Mr. Aguek Ayuen Anyieth and Ms. Ajah Akuien Garang, for their endless support, encouragement, and sacrifices throughout my academic journey. I am equally grateful to my loving wife, Ms. Adut Garang Kuol, for her patience, understanding, and unwavering support during the many months dedicated to this research project.

Finally, I would like to thank the tax inspectors, systems analysts, and software development teams whose practical insights into tax compliance challenges helped bridge the gap between theory and practice, transforming this mathematical framework into a meaningful and potentially impactful operational tool.

To all those who contributed directly or indirectly to the success of this study, I extend my sincere appreciation and gratitude.

TABLE OF CONTENTS

1.0 Introduction 1

1.1 Background 1

1.1.1 Historical Perspective 1

1.1.2 Theoretical Perspective 3

1.1.3 Conceptual Perspective 4

1.1.4 Contextual Perspective 5

1.2 Problem Statement 7

1.3 Purpose of the Study 8

1.4 Objectives 9

1.4.1 Specific Objectives 9

1.4.2 Research Question; The research questions for this study were; 9

1.6 Scope 10

1.7 Significance of the Study 11

2.0 Introduction 12

2.1 Theoretical Review 12

2.2 Conceptual Framework 13

2.4 Related Literature & Research Gap 19

2.4.1 Taxpayer Risk and Revenue Leakage 21

2.4.2 Audit Selection Practices in Tax Administrations 22

2.4.3 Taxpayer Characteristics and Compliance Behaviour 22

2.4.4 Analytical Approaches to Risk-Based Audit Selection 22

2.4.5 Research Gap 23

3.0 Introduction 25

3.1 Study Design 25

3.2 Study Population 26

3.3 Sample Size and Sample Size Determination 26

3.3.1 Sample Size 26
3.3.2 Sample Size Determination 27
3.4 Sampling Technique 28
3.5 Data Collection 28
3.6 Data Collection Instruments 29
3.7 Data Quality Control 30
3.8 Data Collection Procedures 31
3.9 Data Analysis 32
3.10 Implementation of the Framework 33
3.10.1 Visual Implementation 33
3.10.2 Business and Audit Requirement Analysis 33
3.10.3 Data Acquisition 34
3.10.4 Data Preparation & Governance 34
3.10.5 Feature Engineering & Behavioural Profiling 35
3.10.6 Model Development 37
3.10.7 Model Evaluation 39
3.10.8 Risk Classification and Classification 40
3.10.9 System Development 41
3.10.10 Audit Decision Support 42
3.10.11 Continuous Improvement and Maintenance 42
3.10.12 Institutional Stability & Drift Governance 43
3.10.12.1 Model Degradation and Evasion Mechanics 43
3.12 The Retraining Loop Architecture 43
3.11 Ethical Considerations 44
4.1 Introduction 45
4.2 Description of Dataset Profile and Demographics 45
4.2.0 Description of the Raw Dataset Schema 46

4.2.0.1 Column-by-Column Architectural Mapping 46
4.2.1 Distribution of Taxpayers by Primary Transaction Currency 48
4.2.2 Dataset Variables 49
4.2.3 Taxpayer Profile Distribution by Historical Compliance Status 49
4.3 Data Pre-Processing and Behavioural Feature Engineering 50
4.3.1 Data Cleaning and Preprocessing 50
4.3.2 Model Training 51
4.3.3 Model Testing and Evaluation 52
4.3.4 Comparison between the SML system and the legacy manual system ..... **Error! Bookmark not defined.**
4.3.5 Mathematical Resolution of Scale Skewness 53
4.4 Model Performance and Presentation of Results 54
4.4.1 Validation Performance Matrix 54
4.4.2 Confusion Matrix Analysis 55
4.5 Operational Interpretation and Risk Banding 56
4.6 Comparative Analysis: Developed Framework vs. Legacy SSRA Process 57
4.7 Summary 58
5.1 Introduction 60
5.2 Discussions of the Key Findings 60
5.2.1 The Critical Impact of Multi-Currency Feature Normalization 60
5.2.2 Predictive Performance and Classification Integrity 61
5.2.3 Comparative Operational Advantages over Legacy Pipelines 62
5.3 Conclusions 63
5.4 Recommendations 63
5.4.1 Policy and Administrative Recommendations for the SSRA 64
5.4.2 Technical and Implementation Recommendations 64
5.5 Recommendations for Future Research 65

5.6 Summary 66
REFERENCES 67

TABLE OF FIGURES

Figure 1: Showing the conceptual framework between the Independent and Dependent Variables 15
Figure 2: Showing ML application and its potential impacts in SSRA 21
Figure 3: Showing a snippet of the SSRA dataset 29
Figure 4: Showing the steps taken for the development 33
Figure 5: Showing the business and audit requirements 33
Figure 6: Showing the data acquisition steps 34
Figure 7: Showing data preparation steps 34
Figure 8: Showing the process of category encoding 35
Figure 9: Showing feature engineering processes 36
Figure 10: Showing taxpayer aggregation 36
Figure 11: Showing logarithmic transformation 37
Figure 12: Showing model development steps 37
Figure 13: Showing the training of the model 38
Figure 14: Showing model evaluation. 39
Figure 15: Showing the valuation report for the confusion matrix. 39
Figure 16: Showing the risk categories 40
Figure 17: Showing the risk band for high, medium, and low 40
Figure 18: Showing the system development procedure 41
Figure 19: Snippet from the system 42
Figure 20: Showing the continuous improvement steps 42
Figure 21: Snippet of the dataset. 46
Figure 22: Showing the data cleaning and preprocessing implemented in this study 51
Figure 23: Showing model training 52
Figure 24: Showing model training and testing part 53
Figure 25: Showing the performance graph 55
Figure 26: Showing confusion matrix 56
Figure 27: Showing the comparison between the legacy system and new system 58

LIST OF ACRONYMS

AI……..…… Artificial Intelligence

API ……... Application Programming Interface

ATAF ……..……. Africa Tax Forum

BPT …… Business Profit Tax

CIT……..….. Corporate Income Tax

DL……..… Deep Learning

FP…….…….. False Positive

FN…….…….. False Negative

GDP …….…….. Gross Domestic Product

KSH …….……. Kenyan Shilling

ML……..... Machine Learning

NTA …….…….. National Taxpayers Association

OECD……Organization for Economic Co-operation and Development

PIT …….….. Personal Income Tax

SML …….. Supervised Machine Learning

SSP …….. South Sudan Pounds

SSRA ……. South Sudan Revenue Authority

SVM …….……. Support Vector Machine

TIN …… Taxpayer Identification Number

TP…….…….. True Positive

TN……..….. True Negative

VAT……. Value Added Tax

USD …… United States Dollar

URA……. Uganda Revenue Authority

ABSTRACT

Modern tax administrations in developing economies struggle with systemic revenue leakage and severe resource constraints, a challenge frequently compounded by subjective, manual compliance audit selection mechanisms. Within the South Sudan Revenue Authority (SSRA), legacy selection pipelines have historically relied on human discretion and static, rule-based heuristics. This dependency results in low non-compliance detection rates, inefficient inspector resource allocation, and vulnerability to human bias.

To resolve these operational bottlenecks, this dissertation presents the design, development, and validation of a novel, ten-stage supervised machine learning (SML) framework that transitions tax administration from manual, reactive enforcement to proactive, data-driven, risk-based selection. By utilizing historical taxpayer transaction databases, the framework aggregates raw, multi-currency records into unified behavioural profiles indexed by Taxpayer Identification Number (TIN) to engineer diagnostic compliance features. To address the significant scale skewness caused by mixed-currency transactions in South Sudanese Pounds (SSP) and United States Dollars (USD), a natural logarithmic scale correction, $y = log(x + 1)$, is implemented, ensuring classification equity across diverse economic profiles. Classification modeling is executed using a class-balanced Logistic Regression estimator, paired with a standard normal scaling pipeline to prevent data leakage, and operationalized through an enterprise-ready Django and PostgreSQL web interface.

The automated system achieves an overall classification accuracy of 90.27%, with a precision of 0.91 and a robust recall of 0.86 in isolating non-compliant taxpayers. By generating a continuous, real-time risk probability index, the automated framework enables administrative supervisors to dynamically isolate high-risk taxpayers, optimize inspector dispatch, drastically reduce erroneous audit recalls, and maintain a self-evolving model retraining loop that actively mitigates concept drift. Ultimately, this research provides a mathematically rigorous, practically validated blueprint for tax intelligence and automated compliance governance in resource-constrained developing nations. When benchmarked against legacy manual selection processes which historically achieved an accuracy rate of only 60% to 70% and suffered from severe inefficiencies where nearly 30% to 40% of all initiated audit selections had to be recalled due to selection errors; the machine learning framework demonstrates a transformative operational advantage.

CHAPTER ONE: INTRODUCTION

1.0 Introduction

This chapter provides an overview of the study and justifies its necessity. The background was first described using theoretical, conceptual, historical, and contextual viewpoints. The problem statement, study purpose, objectives, research questions, hypotheses, scope, and importance are all included in this chapter. When taken as a whole, these sections provide a clear picture of the problem under investigation and the future course of the study.

1.1 Background

Globally, countries are losing four hundred ninety-two (492) billion dollars to tax abuse and evasion (Mansour, 2024). Frauds such as tax evasion, unaudited tax returns, uncollected fines and penalties, unsolicited loopholes are often the leading causes of these million dollars loses to governments around the world. Tax authorities globally need ways to be able to detect these fraudulent activities in near real-time to be able to save money and promote equitable tax collection. Advancements in the field of Artificial Intelligence (AI) and Machine Learning (ML) have come in handy for the detection of some of these fraudulent activities and tax authorities globally are embracing them. Tax administration uses ML to analyse data, detect fraud, prioritize caseloads and realign resources. ML further automates recommendations, predictions and decisions in tax compliance. Machine Learning and rule-based systems, is widely deployed(Soumaya, 2026)

1.1.1 Historical Perspective

Tax authorities have always had to decide which taxpayers to audit, but the methods used in the past were mostly manual. Earlier approaches relied on officers' judgment, basic risk indicators, or random checks. Although these methods served their purpose, they often struggled to handle large and growing volumes of taxpayer information. In many countries, especially in developed tax systems, researchers began experimenting with statistical and later machine-learning tools to identify risky taxpayers more accurately. These experiments demonstrated that models are able to identify non-compliance patterns that are difficult to identify through manual evaluation.

However, a large portion of this work was completed in settings with robust information systems and extensive digital data histories. In nations with younger tax administrations, like South Sudan, where data availability, filing consistency, and audit capacity present particular

difficulties, there is a dearth of study. This gap in applying supervised learning to such contexts motivates the present study.

This was further stressed by more research on the use of Machine Learning to detect tax fraud in VAT and Corporate Income Tax (CIT) (Vardan Baghdasaryan, 2022). The researches that have been carried out in this field only touch the tip of the iceberg because there are more underlying factors that influence an audit, and there are other variables that can be incorporated into the algorithms to make them more efficient and scalable.

For Chartered Accountants (CAs), accuracy, compliance, and client service are non-negotiable. But, sticking to legacy systems and heavy manual work increases the risk of errors, delays, and lost opportunities. By adopting digital accounting tools, automated reconciliation systems, and cloud-based compliance platforms, firms can save time, reduce stress during peak seasons, and provide better service to clients. The shift may feel challenging initially, but in the long run, it helps CAs work smarter, stay ahead of compliance, and focus on advisory roles that genuinely add value. (Virani, 2025)

(Emma, 2025) states that traditional IT audits often have a limited coverage since manual audits typically rely on statistical sampling, examining only a fraction of the total population. This approach can miss critical issues and provide an incomplete picture of your risk landscape. This justifies the need for having an improved way for conducting audits, both in IT and in Tax Administrations, as new technologies such as ML, Deep Learning (DL) can help identify patterns that would be nearly impossible for a normal auditor to catch, they also help in ensuring that a large chunk of data is processed in a blink of an eye, with near-perfect results reading for dissemination and action upon by the relevant authorities.

Other advanced tax authorities worldwide such as Brazil had around USD 140 billion in assessed taxes waiting for decisions in administrative court tax appeals. It takes about six years for the appeal ruling. Under the AI Litigation Project, Brazil employed supervised machine learning (ML) when distributing groups of similar files to the same officers, in order to increase their speed in administering the file and taking decisions. The first trials, conducted with a sample of 2000 manually labelled files, showed that supervised algorithms can attain sensitivity and specificity of over 80%. (OECD, 2025)

Earlier researchers have conducted a study in this field and used algorithms such as support vector machines (SVMs), Decision Trees, Random Forests as some of the Supervised Machine Learning models (Choudhary, 2025). This approach proved effective, however, some of the

unforeseen factors in the earlier researches carried out is the fact that there are different types of taxes that are paid by businesses, institutions, and individuals; some of these taxes include Business Profit Tax (BPT), Excise Tax, Sales Tax/Value-Added Tax (VAT), Personal Income Tax (PIT) among others; these different types of taxes have different thresholds or caps that are set by the SSRA.

1.1.2 Theoretical Perspective

The study is guided by the Risk-Based Compliance Management Theory. The argument is that tax administrations should concentrate their audit efforts on taxpayers who are more likely to be non-compliant rather than treating all taxpayers equally. It also suggests that risk levels can be identified from behavioural patterns, such as payment histories or filing behaviours.

The Kenyan Revenue Authority (KRA) lost 1.7 trillion Kenyan Shillings (approximately USD 13.1 billion) in contentious and uncollected taxes. This amount includes 524 billion which was reported by the taxman as “erroneous, uncollectable and doubtful debts and 1.16 trillion tax arrears that have been outstanding for more than three years. (Mburu, 2024).

A risk-based approach enables you to identify risks and prioritize them based on potential impact and likelihood, leading to the creation of mitigating controls and policies. Risk-based approach ensures compliance; instead of spending endless hours checking boxes that may not matter, Risk-Based Compliance lets you zero in on critical areas, that if neglected, could cause big headaches. So, this ensures that efforts go where they are most needed. (Standard Fusion, 2025)

(White, 2025) brings a new approach of Regulatory compliance and risk management and defines it as the processes and frameworks organizations implement to ensure they adhere to laws, regulations, guidelines, and specifications relevant to their business operations while simultaneously identifying, assessing, and controlling threats to their capital and earnings. This dual focus requires organizations to both follow external rules and proactively manage potential issues that could impact their ability to achieve business objectives. Regulatory compliance focuses on meeting the requirements set by governmental bodies and industry standards, while risk management addresses the identification, analysis, and response to risk factors that could affect an organization. Together, they form a comprehensive approach to protecting organizations from legal penalties, financial losses, and damage to reputation.
These huge losses in uncollected revenue are some of the reasons as to why a risk-based audit selection system needs to be developed so as to be able to flag and follow up cases as soon as

they arise.

Supervised machine learning can be used to implement this idea. It enables the researcher to develop a model that predicts who is more likely to be high-risk using past taxpayer data.

The theory therefore supports the idea of linking measurable taxpayer characteristics to their likelihood of non-compliance.

1.1.3 Conceptual Perspective

(Shapiro, 1959) defined Machine Learning as a field of study that gives the computers the ability to learn without being explicitly programmed.

According to (Artemenko, 2017), Tax risk is a financial risk suggesting possible unforeseen financial losses caused by the introduction of new tax types changes in tax rules and regulations, like cancelling tax breaks or "tax holidays", increasing tax rates for current taxes, changing the procedure and deadlines for making tax payments and other norms of tax legislation Tax risks are also said to be linked with some uncertainty in achieving the objectives of the region, or the economic entity due to unforeseen negative factors in the taxation process.

This study employs supervised machine learning, which deals with models trained on datasets where the intended output which in this case is the risk label is already known.

(Jakkula) defined support vector machines (SVMs) as a set of related supervised learning methods used for classification and regression. They belong to a family of generalized linear classifiers. In another terms, Support Vector Machine (SVM) is a classification and regression prediction tool that uses machine learning theory to maximize predictive accuracy while automatically avoiding over-fit to the data. Support Vector machines can be defined as systems which use hypothesis space of a linear functions in a high dimensional feature space, trained with a learning algorithm from optimization theory that implements a learning bias derived from statistical learning theory.

(Jain, 2024) defined decision trees as one of the most intuitive and widely-used algorithms for classification and regression tasks. They model decision-making processes in a tree-like structure where each internal node represents a feature (attribute), each branch represents a decision rule, and each leaf node represents the outcome. Decision trees are easy to visualize and interpret, making them an excellent tool for explaining the logic behind predictions.

(Jain, 2024) defined a random forest is essentially a collection (or "forest") of decision trees trained on different subsets of the data. Each tree in the forest is trained independently, and the final prediction is made by aggregating the predictions of all trees, either by taking the majority vote in classification or averaging the predictions in regression.

According to the Namibia Revenue Agency, Business Profit Tax is a type of tax collected from companies and it is based on the net income businesses obtain while exercising their business activity, normally during one business year.

According to Tax Foundation, A value-added tax (VAT) is not a tariff, it is a consumption tax assessed on the value added in each production stage of a good or service. Every business along the value chain receives a tax credit for the VAT already paid. The end consumer does not, making it a tax on final consumption

(Kagan, 2025) defined sales tax as a consumption tax imposed by the government on the sale of goods and services. A conventional sales tax is levied at the point of sale (POS), which is collected by the retailer and passed on to the government. A business may be liable for sales taxes in a given jurisdiction if it has a presence there, which can be a brick-and-mortar location, an employee, or an affiliate, depending on the laws in that jurisdiction.

According to the European Commission, excise duties or excise tax are indirect taxes on the sale or use of specific products such as alcohol, tobacco, energy products and electricity.

ActionAid defined Personal Income Tax (PIT) as a direct tax levied on personal income including wages and salaries, director's fees, dividends, royalties and rental income, amongst others.

A risk-based audit selection system refers to a structured way of ranking taxpayers so that scarce audit resources are used more efficiently. Here, it refers to a simple prototype that uses model outputs to help SSRA identify which taxpayers require deeper review.

These operational definitions will guide how the study variables will be handled.

1.1.4 Contextual Perspective

Revenue Authorities regionally continue to lose millions of dollars annually due to under declarations, tax evasions, and corruption.

In light of these challenges, each of the Revenue Authorities in Africa have their fair share of the losses to tax evasions and as such, regional revenue authorities have come up with some

innovations to reduce under declaration and enhance tax audits in their respective revenue authorities, for example, Uganda Revenue Authority developed a system in 2024 dubbed "Real-time Audit" to examine taxpayer's transactions in real-time, and where irregularities are detected, prompt alerts are sent to the taxpayer on how to address the issue. (The Independent, 2024).

(Kinyanjui, 2025) stated that Kenya lost more than Kenyan Shillings (KSH) 500 billion in revenue in 2024 through unchecked tax waivers, exemptions, and corruption, according to a new report by the National Taxpayers Association (NTA). The amount, which represents 3.3 percent of the country's Gross Domestic Product (GDP), was a sharp increase from the Sh393.6 billion lost in 2022.

Tax Evasions have crippled the economies of Africa, for example, Senegal loses an approximate of USD 57 million to USD 153 million annually to customs under declarations. (Davis, 2024). (Schwikowski, 2025) goes on to further state that an approximate of 88 billion dollars leaves Africa every year in tax evasions.

(Mwangi, 2026) stated that data intelligence is of great importance; to build systems that learn from patterns, flag anomalies and turn raw transactions into timely, actionable revenue intelligence. He went on to further state that the power of machine learning and advanced analytics is not about replacing human judgement, but rather amplifying it. Supervised ML models can score risk across millions of records so auditors focus on the highest-impact cases.

In conclusion, the challenges faced by the South Sudan Revenue Authority make traditional audit selection approaches increasingly ineffective. Limited institutional capacity, a growing informal economy, and rising demands for domestic revenue mobilization constrain auditors' ability to systematically identify high-risk taxpayers using manual or random selection methods. These approaches are not only time-consuming and susceptible to human bias, but they also lack the analytical capacity required to process large volumes of taxpayer data and uncover complex patterns of non-compliance. As transaction volumes continue to increase, reliance on conventional audit practices further reduces the likelihood of detecting high-impact revenue risks in a timely manner.

These limitations highlight the need for a data-driven audit selection mechanism based on machine learning. Machine learning techniques are capable of analysing high-dimensional tax data, learning from historical audit outcomes, and identifying latent risk patterns that are difficult to detect through rule-based or manual analysis. By automating the risk assessment

process, machine learning supports auditors by prioritizing tax payments and taxpayers according to estimated risk levels, thereby improving audit efficiency, consistency, and transparency while retaining human oversight in final audit decisions.

In response to this need, this study proposed a supervised machine learning framework for risk-based audit selection at the South Sudan Revenue Authority. Supervised learning is particularly appropriate in this context because historical audit records provide labelled outcomes that distinguish compliant from non-compliant tax payments. Models such as linear regression–based classifiers are employed to estimate the probability of non-compliance and generate interpretable risk scores that can be directly integrated into audit decision-making processes (Hughes & Riedel, 2025). The contribution of this study lies in the design, training, and evaluation of a context-specific supervised learning framework that aligns with the data constraints, audit practices, and institutional realities of SSRA. By embedding predictive analytics into the audit selection process, the framework aims to enhance early identification of high-risk cases, reduce revenue leakages, and support more effective and evidence-based domestic revenue mobilization.

1.2 Problem Statement

Domestic revenue mobilization remains a major challenge for many African countries, particularly those emerging from conflict and institutional instability (Urama & Ogunleye, 2023). The South Sudan Revenue Authority continued to experience significant revenue leakages due to tax evasion, under declaration, weak compliance enforcement, and administrative inefficiencies. Previous audit practices relied heavily on manual reviews, rule-based checks, and discretionary selection of taxpayers for audit. These approaches limit the Authority's ability to consistently identify high-risk cases, resulting in inefficient use of scarce audit resources and the persistence of undetected non-compliance.

As the volume of taxpayer and transaction data increased, auditors were increasingly constrained by the inability to systematically analyse large datasets in a timely and objective manner. This exposed the audit process to human bias, selective enforcement, and missed revenue opportunities, ultimately weakening fiscal capacity and public confidence in the tax system.

Recent research in tax administration highlights the growing use of data analytics and machine learning to improve audit effectiveness and compliance enforcement (Choudhary, 2025). Studies show that risk-based audit selection models can enhance audit yield by identifying

patterns and anomalies associated with non-compliant taxpayer behaviour. In particular, supervised machine learning approaches have been widely applied to classify taxpayers or transactions using historical audit outcomes, demonstrating improved accuracy and efficiency compared to random or manual audit selection methods. (OECD, 2025)

These approaches have been successfully implemented and studied in revenue authorities with relatively advanced digital infrastructure and well-structured datasets. The literature largely focuses on technical model performance, including predictive accuracy and classification metrics, to demonstrate the effectiveness of machine learning in audit selection. (Soumaya, 2026)

Despite the growing body of research on machine learning in tax administration, there was limited empirical evidence on how supervised machine learning frameworks could be designed and applied within low-capacity, resource-constrained, and post-conflict revenue authorities. Existing studies rarely addressed the institutional, data quality, and operational constraints faced by authorities such as the South Sudan Revenue Authority. Moreover, there was a lack of context-specific frameworks that integrate machine learning outputs into practical audit decision-making processes while maintaining human oversight and governance controls.

This study addressed these gaps by developing a supervised machine learning framework for risk-based audit selection tailored to the South Sudan Revenue Authority. The proposed framework leverages historical tax payment and audit data to classify transactions based on compliance risk and generate risk scores to support audit prioritization. By grounding the framework in local data structures, audit practices, and institutional realities, the study contributes a practical and context-sensitive approach to the literature. The research further evaluates the framework using standard performance metrics to assess its effectiveness as a decision-support tool for enhancing audit efficiency, reducing revenue leakages, and strengthening domestic revenue mobilization at SSRA.

1.3 Purpose of the Study

The purpose of this study was to design a supervised machine learning framework to support audit selection at the South Sudan Revenue Authority. The study was concerned with how tax payments can be screened as they are received in order to identify those that may be fraudulent and require further attention from tax officers. Early detection was important because delayed audits often resulted in missed revenue and weak enforcement. Supervised machine learning was used in this study to classify tax payment records based on past audit outcomes.

Classification was suitable because audit decisions were based on whether a payment complied with set rules and thresholds or not. In practice, a payment was either acceptable or it must be examined further, and there was no meaningful category between these two outcomes; hence making it a binary outcome. The framework developed in this study was intended to assist tax officers in selecting audit cases rather than to replace their role. By narrowing down the number of payments that require attention, the system is expected to help auditors focus on higher-risk cases and make better use of limited audit resources within the South Sudan Revenue Authority.

1.4 Objectives

Main Objectives

The main objective of this study was to design a Supervised Machine Learning Framework using Classification Algorithms to speed up the process of identifying risky tax payments for audit selection.

1.4.1 Specific Objectives

Specific Objectives of this study were;

1. To identify the key requirements for developing and deploying a supervised ML framework for risk-based audit selection.
2. To design and train a Supervised ML framework for risk-based audit selection for the SSRA
3. To evaluate the performance of the trained ML framework using standard metrics such as accuracy, precision, recall, F1-score, and confusion matrix.
4. To deploy a risk-based audit selection framework to be used in the SSRA for fast analysis and selection of fraudulent payments.

1.4.2 Research Question;

The research questions for this study were;

1. What requirements were necessary for the development and deployment of a supervised machine learning framework for risk-based audit selection at the South Sudan Revenue Authority?

2. How can a supervised machine learning framework for risk-based audit selection be designed and trained to fit the operational and data environment of the South Sudan Revenue Authority?
3. How well did the developed supervised machine learning framework perform when evaluated using standard classification metrics such as accuracy, precision, recall, F1-score, and the confusion matrix?
4. How could a trained supervised machine learning framework support fast analysis and effective selection of potentially fraudulent tax payments within the South Sudan Revenue Authority?

1.5 Scope

Content scope

This study focused on the analysis of taxpayer behaviour as captured in administrative data held by the South Sudan Revenue Authority. It covered the application of supervised machine learning techniques for classifying tax payments based on compliance risk and identifying patterns associated with under-declaration and potential fraud. The study further involved the design and development of a model-based risk-based audit selection framework intended to support audit decision-making. The scope excluded policy analysis and focused strictly on data-driven audit selection and model performance evaluation.

Geographical scope

The geographical scope of the study was limited to taxpayers registered and operating under the jurisdiction of the South Sudan Revenue Authority. All data used in the study were derived from SSRA systems and reflected the institutional, legal, and economic context of South Sudan. The findings were therefore specific to the SSRA and might not be directly generalizable to other revenue authorities without contextual adaptation. However, the methodological approach might provide insights applicable to similar revenue administrations.

Sample scope

The study utilized available historical taxpayer and tax payment records extracted from SSRA operational systems for model development and analysis. These records included selected variables relevant to audit risk assessment, subject to data availability and quality constraints. The sample was limited to documented transactions with identifiable audit outcomes suitable

for supervised learning. No new data collection from taxpayers was undertaken, and the analysis was confined to secondary administrative data.

1.6 Significance of the Study

The study was important for several reasons. By identifying high-risk taxpayers in a more open and objective manner, the SSRA might be able to make better use of its limited audit resources. The insights could help management and policy departments identify common compliance issues in various industries, which could guide focused solutions. From an academic standpoint, the work contributed to the scant literature on machine learning in developing-nation tax administration. In practical terms, the suggestions could help the SSRA establish the groundwork for a more organized risk-based auditing procedure.

CHAPTER TWO: LITERATURE REVIEW

2.0 Introduction

This chapter reviewed existing literature related to the study. It covered the theoretical review underpinning risk-based audit selection, the conceptual framework that explained the relationship between the study variables, and a review of related empirical literature organized according to the specific objectives of the study. The chapter provides a foundation for understanding how previous studies and theories informed the current research.

2.1 Theoretical Review

This section expands on the theoretical foundation introduced in Chapter One. The study was guided by the Risk-Based Compliance Management Theory, which underpins modern tax administration practices.

The Risk-Based Compliance Management Theory is commonly associated with tax administration frameworks developed by institutions such as the OECD in the early 2000s. The theory argues that taxpayers differ in their compliance behaviour and that enforcement actions should therefore be differentiated rather than applied uniformly. According to this theory, tax administrations achieve better outcomes when they concentrate audit and enforcement efforts on high-risk taxpayers while minimizing unnecessary interventions for compliant taxpayers.

The theory assumes that non-compliance is not random. Instead, it is influenced by identifiable behavioural, financial, and structural factors such as irregular filing, delayed payments, underreporting of taxable income, and sector-specific risk patterns. By analysing these indicators, tax authorities can estimate the likelihood that a taxpayer poses a compliance risk.

In the context of this study, taxpayer characteristics represented observable indicators of compliance behaviour, while taxpayer risk classification represented the outcome of interest. Supervised machine learning provided a systematic way to operationalize the theory by learning patterns from historical data where compliance outcomes were already known. The models applied statistical learning principles to estimate risk based on past behaviour, consistent with the theory's assumption that risk can be inferred from data.

Therefore, the Risk-Based Compliance Management Theory linked the independent variable (taxpayer characteristics) to the dependent variable (taxpayer risk classification). This theoretical relationship justified the use of supervised machine learning models to support audit selection decisions at the South Sudan Revenue Authority.

2.2 Conceptual Framework

This study looked at how the factors that make a taxpayer who they are affected how they are classified as a risk. The study used computer programs that could learn from the information given to them. It took the ideas that were talked about at the beginning of chapter one and turned them into measurable variables. This helped to guide the analysis of the information. The study of taxpayer-related characteristics was important to understand taxpayer risk classification. The taxpayer-related characteristics were used to learn more about taxpayer risk classification through the use of computer programs.

The taxpayer risk classification was what was being determined in this study. It was the target variable of interest, specifically, whether a taxpayer was considered high risk or low risk when it came to being audited. The taxpayer risk classification shows how likely it is that a taxpayer will not do what they are supposed to do with their taxes, like not reporting all of their income, paying late, or not filing their taxes correctly. The taxpayer risk classification is important because it helps us understand if a taxpayer is going to follow the tax rules or not.

The main independent focus evaluated in this study was taxpayer characteristics. Taxpayer characteristics showed attributes about taxpayers that could be observed and historical occurrences. These attributes were related to how taxpayers behaved and what obligations they had to fulfil. They included factors like how often taxpayers made late payments, if their payments fluctuated significantly, if they received penalties, and if they missed filing deadlines. Taxpayer characteristics are important because they help us understand taxpayer behaviour and taxpayer obligations. These factors also contained important technical indicators, such as when the tax that was expected did not match the tax that was reported. They also contained information about the kind of business, such as what kind of taxes they had to pay (e.g., business profit tax, VAT, sales tax, excise duty, or personal income tax). They incorporated what happened during past audits. The tax risk was inherently reflected in these details, which are elements discussed when considering taxation frameworks broadly and the core operational ideas behind tax enforcement.

Supervised machine learning was utilized as a way to figure out how certain variables about a taxpayer were connected and how much risk they posed. Data where outcomes were already

known was used to train the algorithm. The model looked at this data and found patterns. It used classification algorithms like decision trees and random forests to do this.

The model also used random forests and support vector machines to learn. It learned how attributes about the taxpayer were connected to the risk they posed. Then it used what it had learned to put taxpayers into groups based on the risk they posed. Supervised machine learning helped systematically map taxpayer characteristics to taxpayer risk classification.

The framework operated under the assumption that taxpayer characteristics are critical. If a taxpayer did things like execute highly volatile transactions, or if the tax they said they owed was very different from what was expected, or if they were penalized often, then the framework categorized them as high risk. On the other hand, if a taxpayer always followed the rules, then the framework categorized them as low risk. The taxpayer characteristics, like paying on time, helped decide if the taxpayer was high risk or low risk. The framework used the taxpayer characteristics to compute the risk score for the taxpayer.

The result of this process was a prioritized list that showed which taxpayers were more likely to be non-compliant with their taxes. The taxpayers were classified into groups based on how likely it was that they were not following the tax rules. This list helped the people in charge of making decisions about who should be audited. It told them which taxpayers they should look at first so they could make sure everything was fair and they were using their time wisely when choosing who to audit. The audit selection process became more efficient and objective because of this framework.

For our study, we used Logistic Regression because of the output of the research is a probability-based binary classification problem and the model provides probability-based outputs that are rule-based. Logistic Regression is also computationally efficient, relatively transparent and light weight. This makes it ideal for resource-constrained institutions like the South Sudan Revenue Authority.

In summary, the conceptual framework linked taxpayer characteristics (independent variable) to taxpayer risk classification (dependent variable) through supervised machine learning

models. This linkage provided the foundation for the empirical analysis and guided both model development and interpretation in the study.

Figure 2.1: Conceptual Framework for Risk-Based Audit Selection

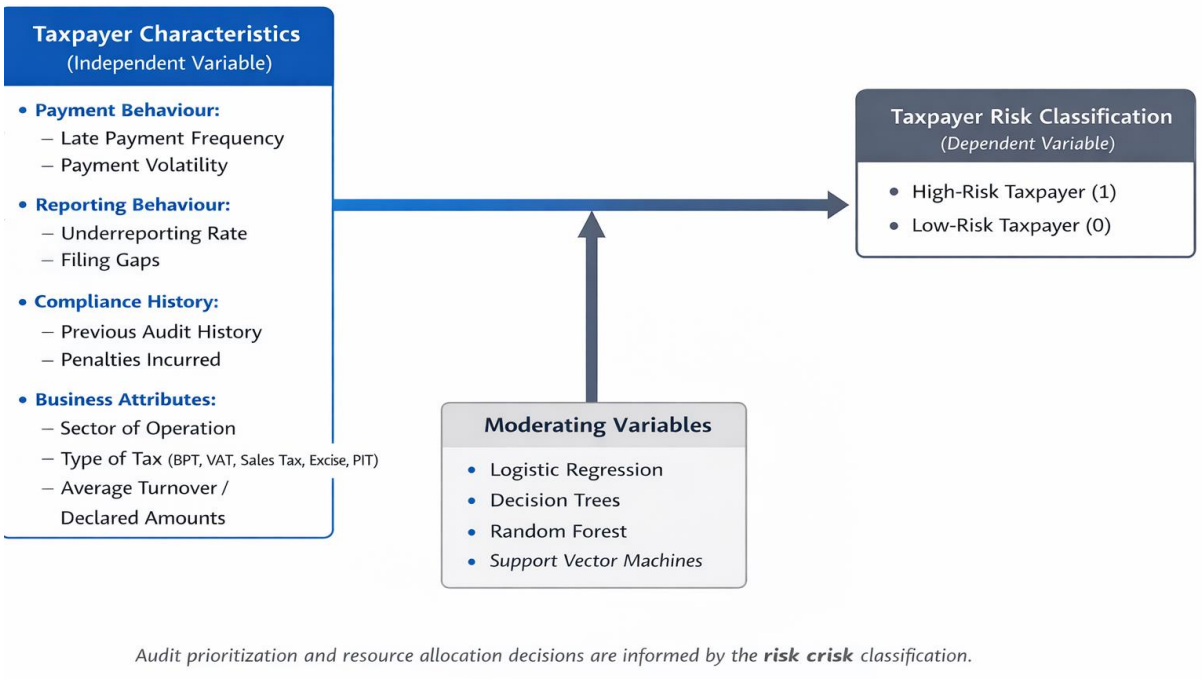


Figure 1: Showing the conceptual framework between the Independent and Dependent Variables

Independent Variable: Taxpayer Characteristics.

In this study, taxpayer Characteristics were the independent variables and this refered to the patterns such as payment behaviour, reporting behaviour, compliance history, business attributes. These attributes of taxpayer characteristics constituted how well a taxpayer will be classified, whether he/she be a high-risk taxpayer or a low-risk taxpayer.

Dependent Variable: Taxpayer Risk Classification.

In this study, risk classification was the dependent variable because the outcome of this variable was dependent on the payment characteristics of the taxpayer.

Mathematical Formulation of the Supervised ML Framework for Risk-Based Audit Selection

1. Problem Definition in Mathematical Terms

In this study, the objective was to classify tax payments into risky (fraudulent) or non-risky (compliant) categories.

Let:

$$D = \{(xi, yi)\}_{i=1}^{N} \quad (2.1)$$

Where $xi \in R^n$ represents a feature vector describing a tax payment (e.g. declared amount, payment history, filing frequency, taxpayer category, sector, exemptions). $yi \in \{0,1\}$ is the class label and zero (0) represents a compliant payment while one (1) represents a potentially fraudulent payment.

This formulation framed audit selection as a binary classification problem, which justifies the use of supervised learning models.

2. Supervised Classification Function

The supervised ML model learns a function:

$$f(x): R^n \rightarrow \{0,1\} \quad (2.2)$$

Such that:

$$y_i = f(x_i) \quad (2.3)$$

Where $y\hat{}i$ is the predicted audit risk class for tax payment iii

The goal of training was to minimize misclassification between predicted labels $y\hat{}$ and true labels y.

3. Loss Function Optimization

Model training was performed by minimizing a loss function LLL, commonly binary cross-entropy for classification tasks:

$$L(y, y\hat{}) = -[ylog(y\hat{}) + (1-y)log(1-y\hat{})] \quad (2.4)$$

The optimization objective is:

$$\min_{\theta} \sum_{i=1}^{N} L\big(y_i, f(x_i; \theta)\big) \quad (2.5)$$

Where θ represents model parameters (weights and biases) and the loss penalizes incorrect risk classification

This ensures the model learns patterns that distinguish fraudulent from compliant tax payments.

4. Parameter Update Rule (Learning Process)

Model parameters are updated iteratively using gradient descent:

$$\theta^{(t+1)} = \theta^{(t)} - \frac{\alpha}{(\partial L)(\partial \theta)} \quad (2.6)$$

Where α is the learning rate. A stable learning rate ensures convergence without oscillation

A poorly selected learning rate may result in unstable risk scores, reducing the reliability of audit selection decisions at SSRA.

5. Decision Threshold for Audit Selection

Instead of directly using class labels, the model produces a risk probability score:

$$p(y = 1 \mid x)$$

An audit was triggered when:

$$p(y = 1 \mid x) \geq \tau$$

Where τ is the audit risk threshold defined by SSRA policy

This allowed the Authority to control audit volume, prioritize high-risk cases and adapt to capacity constraints

6. Effect of Label Noise in Tax Audit Data

Due to imperfect audits and human judgment, some labels may be incorrect:

$$P(y = y\hat{}) \uparrow \Rightarrow L \uparrow$$

This reflected the reality that not all fraud is detected and some compliant taxpayers may be wrongly flagged

Your research acknowledges this limitation and evaluates model robustness accordingly.

7. Performance Evaluation Metrics

The trained model was evaluated using a confusion matrix:

Actual / Predicted	**Fraud (1)**	**Compliant (0)**
Fraud (1)	TP	FN
Compliant (0)	FP	TN

From which the following metrics are computed:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP} \qquad 2.7$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

This machine learning approach tackled audit selection by treating it as a straightforward yes-or-no question: *Is a tax payment compliant, or is it suspicious?* Every transaction was broken down into a set of features, details about the taxpayer and how they usually paid. The goal was to build a model that sorted out which payments were legitimate and which ones deserved a closer look, using past, labelled data as a guide. Training the model was centered around minimizing mistakes. It tweaked its own parameters through gradient-based methods, and the learning rate was highly critical, set it wrong and the whole system could wobble or stall.

Instead of just spitting out a static "yes" or "no," the model gave each payment a risk score. This way, the South Sudan Revenue Authority could set its own audit cut-offs depending on how many cases it could actually handle and what its current operational priorities were. Real-world tax data was imperfect; some historical audit records were missing, and labels could be off. The study did not ignore this. It measured how well the model performed using standard metrics like accuracy, precision, recall, F1-score, and confusion matrix analysis. The idea was to ensure that the audit selection process stayed reliable and easy to understand, even when dealing with messy administrative data.

2.3 Related Literature & Research Gap

This section reviewed existing literature related to taxpayer risk assessment, audit selection practices, and the use of analytical approaches in tax administration. The review is organized in line with the study objectives.

Data quality challenges have also been acknowledged in recent literature. UNDP (2024) and the World Bank (2024) highlight that many African revenue authorities face issues related to incomplete records, inconsistent audit outcomes, and weak data governance structures. While those studies recognized the importance of data quality, they did not empirically examine how such limitations affect the performance and reliability of supervised machine learning models used for audit selection.

Recent studies have examined the application of data analytics and machine learning techniques in tax administration to improve compliance and fraud detection. For example, the (OECD & Bank, 2024; 2025) report that advanced analytics and machine learning models have been successfully deployed in several developed tax administrations to enhance audit targeting and compliance monitoring. These studies demonstrated improved audit efficiency and revenue outcomes in environments with mature digital infrastructure and well-integrated data systems. However, the reports largely focused on advanced economies and did not provide empirical

evidence on how such approaches could be adapted to fragile and resource-constrained revenue authorities, limiting their applicability to contexts such as the South Sudan Revenue Authority.

(Spinelli, 2025) proposed a machine learning-based approach for detecting tax fraud using historical taxpayer data and post-audit outcomes. Their study showed that supervised classification models can achieve high predictive accuracy in identifying fraudulent behaviour. Nevertheless, the proposed approach was applied primarily to post-audit fraud detection and did not address risk-based audit selection at the point of tax payment, where early identification of high-risk transactions could prevent revenue loss before it occurs.

Similarly, (Martinez, 2025) evaluates the use of machine learning models to support tax compliance analysis and fraud investigation. While that study demonstrated the effectiveness of supervised learning techniques in identifying non-compliant taxpayers, it placed limited emphasis on real-time or near real-time audit selection. The research did not explore how machine learning outputs could be operationalized to support timely audit decisions within constrained institutional environments.

Other studies, such as those reported in Frontiers in (OECD, 2025), focus on evaluating machine learning models using technical performance metrics including accuracy, precision, and recall. These studies provided valuable insights into model performance but gave limited attention to the integration of machine learning outputs into day-to-day audit decision-making processes. As a result, there remained a gap in understanding how such models could be used as decision-support tools that enhance, rather than replace, professional judgment and institutional accountability.

Finally, (ATAF, 2025) discusses policy-level initiatives promoting the adoption of artificial intelligence and advanced analytics within African revenue authorities. Although the report outlined strategic directions and potential benefits, it lacked peer-reviewed empirical studies demonstrating the design, implementation, and evaluation of supervised machine learning frameworks for audit selection in African contexts. In particular, there was no documented study that developed a context-specific supervised machine learning framework tailored to the institutional and operational realities of the South Sudan Revenue Authority.

In response to these gaps, this study developed and evaluated a supervised machine learning framework for risk-based audit selection designed specifically for the South Sudan Revenue Authority. By focusing on early risk identification, operational integration, and local data

constraints, the study contributed empirical evidence and a practical framework that extended existing literature on machine learning in tax administration.

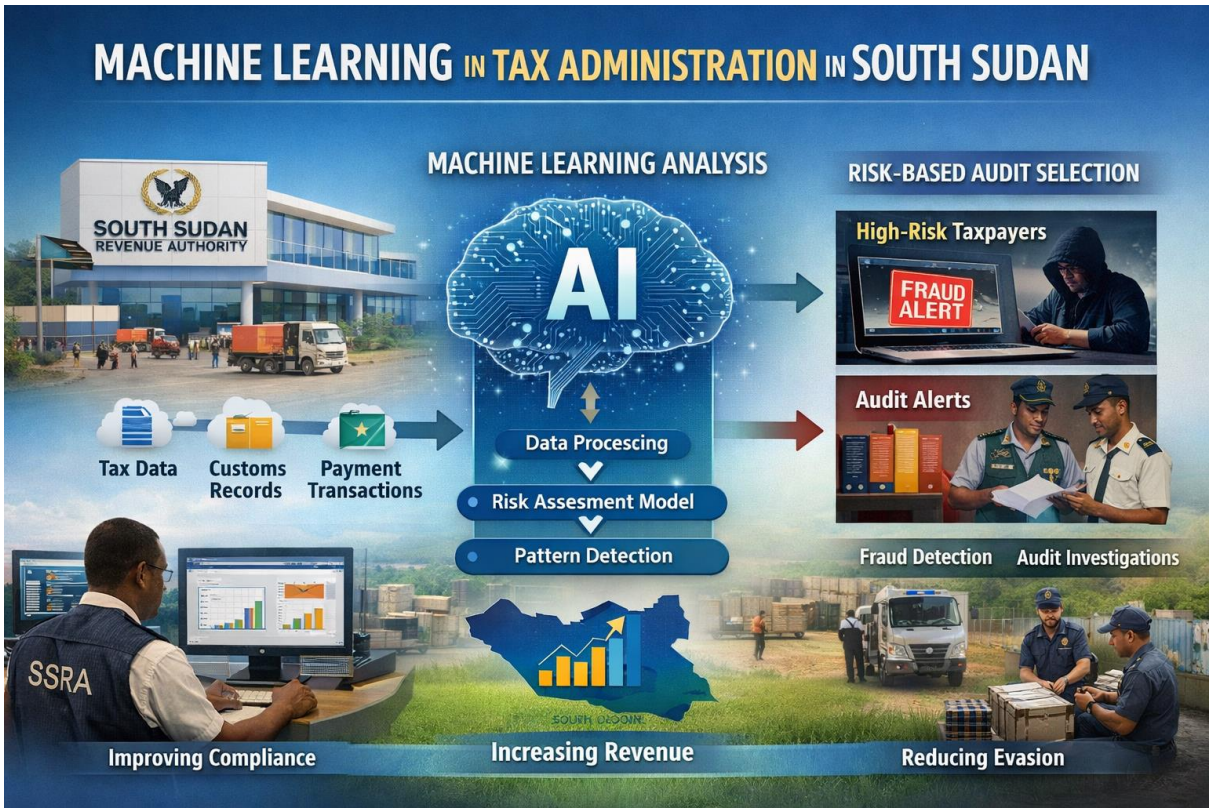


Figure 2: Showing ML application and its potential impacts in SSRA

2.3.1 Taxpayer Risk and Revenue Leakage

Several studies have identified taxpayer risk as a major contributor to revenue leakage in tax administrations, particularly in developing economies. According to (Artemenko, 2017), tax risk arises from uncertainty in tax regulations, taxpayer behaviour, and administrative enforcement, often leading to unanticipated revenue losses. Weak enforcement mechanisms and inconsistent audit practices further compound this risk.

Empirical studies indicate that manual or discretionary audit selection systems tend to focus repeatedly on visible or compliant taxpayers while missing high-risk cases. (Bird & Zolt, 2008) observe that such practices undermine horizontal equity, as compliant taxpayers bear a disproportionate compliance burden. This results in declining voluntary compliance over time, as compliant taxpayers perceive the system as unfair.

These findings related directly to the present study, which sought to address revenue leakage by improving how taxpayer risk was identified and acted upon.

2.3.2 Audit Selection Practices in Tax Administrations

Audit selection is widely recognized as a critical function in tax administration. Traditional audit selection methods rely on random selection, officer judgment, or historical patterns. While these methods are simple to implement, (OECD, 2010) notes that they are often inefficient and susceptible to bias, especially where staff capacity and data quality are limited.

Research conducted by (Khwaja, Awasthi, & Loeprick, 2011) shows that risk-based audit selection leads to higher audit yields compared to blanket or random audits. By focusing on taxpayers with indicators of non-compliance, tax authorities can achieve better revenue outcomes while reducing unnecessary audits on compliant entities.

The relevance of this literature to the completed study lay in its emphasis on structured and evidence-based audit selection as a response to limited administrative resources.

2.3.3 Taxpayer Characteristics and Compliance Behaviour

Existing literature has established a strong relationship between taxpayer characteristics and compliance behaviour. Factors such as business sector, tax type, payment history, filing consistency, and prior enforcement actions have been shown to influence compliance outcomes.

According to (Kirchler, 2007), compliance behaviour varies significantly across taxpayer categories, with certain sectors exhibiting higher risk due to cash-based operations, informal practices, or regulatory complexity. Similarly, (Torgler, 2011) argues that past behaviour is a reliable indicator of future compliance, particularly where enforcement is weak.

These studies supported the inclusion of taxpayer characteristics as key inputs in assessing risk, forming the empirical basis for the independent variable handled in this study.

2.3.4 Analytical Approaches to Risk-Based Audit Selection

Recent literature indicates a gradual shift toward the use of data-driven approaches in audit selection. While earlier studies relied on statistical scoring methods, more recent work explores classification techniques to improve accuracy in identifying high-risk taxpayers.

For example, (Jain, 2024) demonstrates that structured decision rules derived from historical audit outcomes can improve consistency in audit decisions. Similarly, (OECD, 2019) highlights that analytical tools, when used appropriately, enhance transparency and accountability in audit selection, provided they are aligned with institutional capacity.

Rather than replacing human judgment, these approaches are presented as decision-support mechanisms. This aligned with the current study's focus on having developed an interpretable framework suitable for practical institutional use.

2.3.5 Research Gap

Paper	Method	Gap
(Choudhary, 2025) Hybrid Approach to Tax Fraud Detection using Machine Learning by Harsh Choudhary	SVM, Random Forests, Decision Trees.	SVMs are good but require a lot of data to gain optimal performance. The paper does not have any topic to do with Tax Types and Industry Type differences for taxpayers.
(Vardan Baghdasaryan, 2022) Improving Tax Audit Efficiency Using Machine Learning: The Role of Taxpayer's Network Data in Fraud Detection. Vardan Baghdasaryan, Hrant Davtyan, Arsine Sarikyan & Zaruhi Navasardyan (2022)	Supervised and Unsupervised ML	Focuses on VAT (Value-Added Tax) and CIT (Corporate Income Tax). Low range of application due to inadequate data from Armenia Revenue Service.
(Hassan, 2025) The use of artificial intelligence techniques and their impact on the independence of the internal auditor and the quality of electronic financial reports.	Supervised ML	The research focuses on Electronic Fiscal Records only.

2.4 Chapter Summary

This chapter reviewed the theoretical, conceptual, and empirical literature relevant to the development of a supervised machine learning framework for risk-based audit selection at the South Sudan Revenue Authority (SSRA). The review established that the increasing volume and complexity of taxpayer and payment data have created a need for more systematic and data-driven approaches to tax compliance risk assessment. The theoretical review provided the foundation for understanding how machine learning can be applied to identify patterns within historical taxpayer data and use these patterns to support the classification and prioritization of potentially high-risk taxpayers.

The conceptual review established that effective risk-based audit selection depends on the availability of relevant taxpayer data, appropriate data preparation, meaningful feature engineering, suitable supervised learning algorithms, and reliable evaluation mechanisms. Taxpayer payment characteristics such as payment frequency, payment amounts, payment gaps, payment volatility, and currency usage were identified as potentially useful indicators for constructing taxpayer behavioural profiles. The review also demonstrated that machine learning-based audit selection should be regarded as a decision-support mechanism rather than a replacement for professional audit judgement.

The empirical literature demonstrated the increasing application of supervised machine learning techniques in areas involving fraud detection, tax compliance, risk assessment, and audit selection. Previous studies generally reported that machine learning could improve the identification and prioritization of high-risk cases when compared with purely manual or rule-based approaches. However, the reviewed studies also highlighted challenges associated with data quality, class imbalance, feature selection, model interpretability, changing behavioural patterns, and the transferability of models between different institutional environments.

Despite the growing body of research on machine learning for fraud detection and tax compliance, the literature revealed a limited number of studies addressing the specific operational and data environment of the South Sudan Revenue Authority. In particular, there was limited evidence of an end-to-end supervised machine learning framework that moved beyond theoretical model development to include taxpayer data preparation, behavioural feature engineering, model training, quantitative evaluation, risk classification, and practical deployment within a revenue administration environment. This gap provided the justification for the present study.

The literature review therefore informed the methodological approach adopted in Chapter Three. The identified requirements and gaps guided the selection of a supervised machine learning approach, the preparation and aggregation of taxpayer payment data, the engineering of taxpayer behavioural features, the selection and training of the classification model, and the use of accuracy, precision, recall, F1-score, and confusion matrix analysis for model evaluation. The literature also informed the decision to develop the model as part of an operational framework capable of supporting practical audit-risk prioritization. Consequently, Chapter Three presents the research methodology and implementation procedures used to develop, train, evaluate, and deploy the proposed supervised machine learning framework for risk-based audit selection at SSRA.

CHAPTER THREE: METHODOLOGY

3.0 Introduction

This chapter outlines the methodology that was used to carry out the study on the development of a supervised machine learning framework for risk-based audit selection at SSRA. It outlines the research design, study population, sample size and its determination, sampling techniques used. The methodology of this research is designed to be effective in delivering the most optimum solution for risk-based audit solution. This includes looking into the applicability of all Supervised Machine Learning models to achieve the most optimum outcome. This includes data preparation, training of the models, testing of the models, and evaluation.

3.1 Study Design

This study adopted a mixed-methods research design by integrating quantitative data analysis with qualitative insights. The quantitative component was dominant and involved the use of numerical taxpayer data to identify patterns, relationships, and outcomes related to audit risk classification. This included the development, training, and evaluation of supervised machine learning models, where different algorithms and parameter settings were systematically tested and compared to assess their performance in identifying high-risk tax payments.

The study focused on measurable taxpayer behaviour such as filing consistency, payment patterns, compliance history, and tax type information. These variables were transformed into numerical features and used as inputs to predictive models within an experimental framework, allowing for controlled testing and evaluation of model performance.

The qualitative component was incorporated through interviews with selected audit staff at the South Sudan Revenue Authority. The interviews were necessary because there is no documented threshold for flagging taxpayers, it is all known knowledge that isn't documented, this ensure that we fine-tuned the thresholds of the model into what is already known by the auditors. These interviews provided insights into existing audit practices, risk indicators, and institutional processes, which informed feature selection, support interpretation of model outputs, and ensured that the developed framework aligned with real-world audit decision-making.

The implementation was carried out using the Python programming language and supporting libraries such as pandas, NumPy, scikit-learn, and matplotlib. By combining empirical data

analysis with practical insights, the study provided a framework that is both technically robust and operationally relevant for risk-based audit selection.

3.2 Study Population

The population of the study consisted of all taxpayers registered under the SSRA. These taxpayers formed the relevant population because they were subject to tax assessment, payment obligations, and audit procedures administered by SSRA. As such, they directly experienced the outcomes of audit selection decisions.

In addition to taxpayers, the study also considered a small population of SSRA staff, particularly officers from the audit department and other relevant units. These officers were included not as statistical subjects, but as key informants who could provide insight into existing audit practices and institutional processes.

With growing concerns around data privacy, having a private dataset ensures control over data handling and security. This is particularly crucial when dealing with sensitive information. (Glenn, 2023)

3.3 Sample Size and Sample Size Determination

3.3.1 Sample Size

The sample size for this study consisted of a selected subset of historical taxpayer and tax payment records obtained from the South Sudan Revenue Authority's administrative systems. These records were systematically prepared to support the training, optimization, and evaluation of the proposed supervised machine learning framework for risk-based audit selection. The dataset included only transactions with sufficient and reliable information to support model learning and outcome labelling.

To ensure robust model development and unbiased performance assessment, the dataset was partitioned into three mutually exclusive subsets. The training set, comprising approximately 80 percent of the data, was used to train the supervised learning models and enable the identification of patterns associated with compliant and non-compliant tax payments. The validation set, comprising approximately 10 percent of the data, was used during model development to tune hyperparameters, assess model stability, and reduce the risk of overfitting. The testing set, comprising the remaining 10 percent, was reserved exclusively for evaluating the final performance of the trained models on previously unseen data.

This data partitioning strategy followed established best practices in supervised machine learning and predictive analytics. It ensured that model performance metrics such as accuracy, precision, recall, and F1-score were estimated objectively, thereby providing a reliable assessment of the framework's effectiveness in supporting risk-based audit selection at the South Sudan Revenue Authority.

3.3.2 Sample Size Determination

The sample size for this study was determined by the total number of historical taxpayer and tax payment records available within the South Sudan Revenue Authority's administrative systems that met the data quality requirements for supervised learning. Unlike survey-based studies that rely on probabilistic sample size formulas, machine learning studies determine sample size empirically based on data availability, feature completeness, class distribution, and the requirement to ensure sufficient representation of both compliant and non-compliant audit outcomes for reliable model training and generalization.

In this study, the dataset was partitioned into subsets to support supervised model development and unbiased performance evaluation. Approximately 80 percent of the available records were allocated to the training set and were used to learn model parameters and identify patterns associated with audit risk. The remaining portion of the data was reserved for model evaluation, with a validation process applied during training to support hyperparameter tuning, monitor convergence, and reduce the risk of overfitting. A separate testing set was used for final performance assessment on previously unseen records. This data partitioning strategy was consistent with established best practices in supervised classification and predictive analytics, where a larger training proportion is recommended to enhance model learning while preserving sufficient data for objective evaluation.

Given the administrative nature of tax data, issues such as class imbalance between compliant and non-compliant cases were addressed through appropriate data preprocessing and resampling techniques. These approaches helped ensure that the supervised learning models were not biased toward the majority class and were capable of reliably identifying high-risk cases. Overall, the adopted sample size determination strategy ensured that the dataset was adequate and well-structured to support the development, optimization, and evaluation of a supervised machine learning framework for risk-based audit selection at the South Sudan Revenue Authority.

3.4 Sampling Technique

This study employed a non-probability purposive sampling technique. Purposive sampling was appropriate because the study did not involve direct engagement with human subjects, but instead relied on pre-existing administrative taxpayer and tax payment records obtained from the South Sudan Revenue Authority. The dataset was selected based on specific inclusion criteria that were relevant to risk-based audit selection and supervised machine learning model development.

Only tax records that contained complete and usable information on key variables such as filing behaviour, payment patterns, tax type, and historical audit outcomes were included in the study. Records with missing critical fields, inconsistent audit labels, duplicate entries, or obvious data errors were excluded from the analysis. This targeted selection ensured that the data used were suitable for supervised learning, where reliable input features and clearly defined outcome labels are essential for effective model training and evaluation.

The use of purposive sampling allowed the study to focus on records that were most relevant to audit risk classification, rather than attempting to generalize across all available data regardless of quality. This approach was consistent with established practices in machine learning research using administrative and transactional data, where data suitability and labelling accuracy are prioritized to ensure valid and reliable model performance. Through this sampling technique, the study ensured that the resulting dataset supported the development, validation, and testing of a supervised machine learning framework for risk-based audit selection at the South Sudan Revenue Authority.

3.5 Data Collection

This study adopted a secondary data collection method. No primary data was collected directly from taxpayers or tax officers. Instead, the study utilized historical administrative data obtained from the South Sudan Revenue Authority's eTax system; this data was obtained by the researcher with permission from the institution. The dataset consisted of previously recorded taxpayer and tax payment information that was routinely collected for compliance and revenue administration purposes.

	PAYMENT DATE	REFERENCE NUMBER	INVOICE NUMBER	TYPE	TIN	NAME	CURRENCY	AMOUNT PAID
0	31-12-2025	ETAXBQRHGGH366B8	LNKZVKD	tax_return	B20211102VDMGM4	UGANDA NATIONAL AIRLINES COMPANY LIMITED	USD	360.0
1	31-12-2025	ETAXD3HQN3PB4B4Q	EZEBXEN	tax_return	B20211102VDMGM4	UGANDA NATIONAL AIRLINES COMPANY LIMITED	USD	360.0
2	31-12-2025	ETAXRRPMLDJDBFQR	NGKMNKG	tax_return	B20211102VDMGM4	UGANDA NATIONAL AIRLINES COMPANY LIMITED	USD	360.0
3	31-12-2025	ETAXD82QJRBLN4FJ	LNKQQMV	tax_return	B20220804VWX8LX	GREEN SOM GENERAL TRADING COMPANY LIMITED	SSP	1161000.0
4	31-12-2025	EXVXWN	KDGWZW	tax_return	B20220804VWX8LX	GREEN SON GENERAL TRADING COMPANY LIMITED	SSP	34000.0

Figure 3: Showing a snippet of the SSRA dataset

The use of secondary data was appropriate for this study for several reasons. First, it ensured ethical compliance by avoiding direct interaction with individuals and minimizing privacy and confidentiality concerns, as the analysis was conducted on anonymized records. Second, secondary data enabled reproducibility and consistency, as the models could be trained and evaluated using stable and well-defined datasets. Third, the availability of large volumes of historical tax data provided a suitable foundation for supervised machine learning, which requires sufficient data to learn patterns associated with compliant and non-compliant behaviour.

By relying on existing administrative records, the study leveraged real-world operational data, ensuring that the developed supervised machine learning framework reflected actual taxpayer behaviour and audit outcomes. This approach supported the development of a practical and evidence-based risk-based audit selection model that could be integrated into the operational environment of the South Sudan Revenue Authority.

3.6 Data Collection Instruments

Given the computational and data-driven nature of this study, software tools and analytical frameworks were used as the primary data collection instruments. These tools facilitated data extraction, preprocessing, model development, and performance evaluation within a supervised machine learning environment. The key instruments used in this study included:

1. Internal Excel Reports**:** These were downloaded by the researcher and contained past payment data of taxpayers.

2. Python Programming Language: This served as the core tool for data handling, web scraping, preprocessing, feature engineering, and the implementation of supervised machine learning algorithms.
3. Pandas and NumPy Libraries: These were used for data cleaning, transformation, and numerical computation on large-scale taxpayer and tax payment datasets.
4. Scikit-learn: This library was employed for developing supervised classification models, model training and testing, hyperparameter tuning, and evaluation using standard performance metrics such as accuracy, precision, recall, F1-score, and the confusion matrix.
5. Matplotlib and Related Visualization Libraries: These tools were used to visualize data distributions, model performance, and classification outcomes to support interpretation and analysis.

No physical measurement instruments, surveys, or questionnaires were used in this study. Instead, the historical taxpayer records obtained from the South Sudan Revenue Authority constituted the primary data source and served as direct inputs into the computational tools. These instruments collectively enabled the systematic and reproducible development of a supervised machine learning framework for risk-based audit selection.

3.7 Data Quality Control

Several measures were implemented in this study to ensure data quality, model reliability, and the validity of the supervised machine learning results:

1. Data Preprocessing: All taxpayer and tax payment records were cleaned and standardized prior to analysis. This included handling missing values, correcting inconsistent entries, removing duplicates, and transforming relevant variables into suitable numerical formats to ensure consistency across records.
2. Feature Verification and Encoding: Key variables such as filing frequency, payment amounts, tax type, industry classification, and audit outcomes were verified for accuracy and completeness. Categorical variables were appropriately encoded to ensure correct representation within the supervised learning models.
3. Class Balance Assessment: The distribution of compliant and non-compliant audit outcomes was examined to identify potential class imbalance. Where necessary,

appropriate resampling or weighting techniques were applied to reduce bias and improve the model's ability to detect high-risk cases.

4. Validation and Model Monitoring: Model performance was monitored during training using a validation dataset to assess convergence and generalization. This process supported hyperparameter tuning and helped reduce the risk of overfitting to the training data.
5. Standardized Performance Metrics: The supervised learning models were evaluated using widely accepted classification metrics, including accuracy, precision, recall, F1-score, and the confusion matrix. The use of standardized metrics ensured objective performance assessment and allowed for a clear comparison with related studies in tax compliance analytics.

3.8 Data Collection Procedures

The data collection process for this study followed a structured and systematic computational workflow designed to support reliable model development and evaluation.

1. Data Acquisition: Historical taxpayer and tax payment records were extracted from the South Sudan Revenue Authority's administrative systems in accordance with institutional data access and confidentiality guidelines.
2. Data Organization: The extracted records were organized into structured datasets based on relevant variables such as filing behaviour, payment information, tax type, industry classification, and historical audit outcomes. The data were prepared in formats suitable for supervised machine learning analysis.
3. Data Preprocessing: The dataset underwent preprocessing procedures, including data cleaning, handling of missing values, feature encoding, and normalization where required. These steps ensured consistency and suitability of the data for model training.
4. Dataset Splitting**:** The processed dataset was partitioned into training, validation, and testing subsets, with approximately 80 percent allocated for model training, 10 percent for validation and model tuning, and 10 percent reserved for final evaluation.
5. Model Training and Evaluation: Supervised machine learning models were trained using the training dataset and optimized through validation-based monitoring and hyperparameter tuning. The final model performance was evaluated using the testing dataset to assess predictive accuracy and audit risk classification effectiveness.

This structured procedure ensured methodological consistency, reproducibility, and rigor in the development of a supervised machine learning framework for risk-based audit selection at the South Sudan Revenue Authority.

3.9 Data Analysis

Data analysis in this study involved a combination of quantitative statistical analysis and computational model evaluation techniques tailored to supervised machine learning–based audit risk classification. The analysis was conducted using the Python programming language with established data science and machine learning libraries, including pandas, NumPy, scikit-learn, and Matplotlib.

The analysis began with exploratory data analysis to examine key characteristics of the taxpayer and tax payment dataset. Descriptive statistics were used to assess variable distributions, identify missing values, detect outliers, and examine class distribution between compliant and non-compliant cases. This preliminary analysis supported informed preprocessing decisions and helped identify potential class imbalance issues that could affect model performance.

The core analytical process focused on training and evaluating supervised classification models for risk-based audit selection. The dataset was partitioned into training, validation, and testing subsets based on predefined proportions. During model training, learning behaviour was monitored using validation-based performance measures to assess convergence, model stability, and the risk of overfitting.

Model performance was quantitatively evaluated using standard classification metrics, including accuracy, precision, recall, F1-score, and confusion matrix analysis. These metrics provided insight into both overall predictive performance and the model's ability to correctly identify high-risk tax payments. Where appropriate, comparative analysis was conducted by benchmarking the results of the proposed framework against baseline models or findings reported in recent tax compliance and machine learning literature.

The results of the analysis were presented using tables, charts, and graphical visualizations to support clear interpretation and objective assessment of the supervised machine learning framework's effectiveness in improving audit selection at the South Sudan Revenue Authority.

3.10 Implementation of the Framework

3.10.1 Visual Implementation

To successfully transition an administrative entity from legacy manual reviews to automated risk scoring, execution teams had to follow a strict, sequential pipeline. Below is a detailed flowchart showing how raw transactional records were systematically processed into long-term institutional intelligence.

https://nra-etax.online/gmt/simulation_sound_box/gmt-audit-sound-box.html



Figure 4: Showing the steps taken for the development of a RISK-BASED Audit framework

3.10.2 Business and Audit Requirement Analysis



Figure 5: Showing the business and audit requirements

The purpose of this was to formally establish the compliance problem that the ML system would solve. The operational inputs included the legacy audit policy papers, revenue authority compliance objectives, manual selection rules, and stakeholder functional requirements. The SSRA identified a heavy reliance on manual file reviews, which skewed inspector resources.

The technical objective was defined as engineering an automated system capable of uncovering hidden evasion markers. The outputs of this phase were; documented ML audit goals, standardized risk identification criteria, and operational technical specification drafts.

3.10.3 Data Acquisition

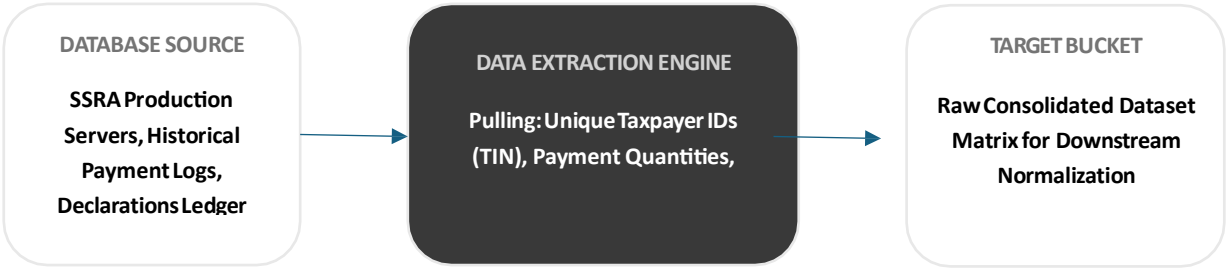


Figure 6: Showing the data acquisition steps

Core Purpose: To gather and extract historical taxpayer transaction registries to construct the database backbone required for model development.

Operational Inputs: Taxpayer payment ledgers, payment execution timestamps, raw transaction values, currency identifiers (e.g., SSP, USD), and unique Taxpayer Identification Numbers (TIN).

Stage Activities & Mechanics: Engineers connected directly to production databases to extract core transaction logs. Before formatting began, datasets were validated to ensure data completeness and check system accessibility ratios.

SSRA Case Implementation: Information technology teams isolated structured ledger sets directly from administrative servers. The resulting table contained historical transaction lines alongside pre-existing risk verification status markers used during system training.

3.10.4 Data Preparation & Governance

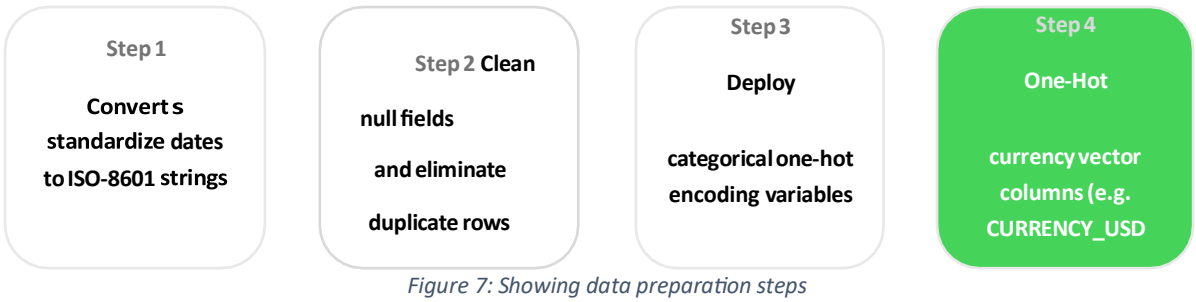


Figure 7: Showing data preparation steps

Core Purpose: The core purpose of this stage was to fix data quality anomalies, handle missing information blocks, and vectorize variables into machine-readable types.

Operational Inputs: The operational input used for this process was the raw extracted taxpayer transaction dataset.

Stage Activities & Mechanics: This stage executed strict duplicate removal, normalized formatting by casting diverse dates to ISO-8601 standards, and addressed null values. Categorical text columns, such as currency codes, were transformed into boolean indicators using one-hot vectorization methods.

PYTHON SNIPPET: CATEGORY ENCODING

```
# Executing categorical transformation for currency variables
df = pd.get_dummies(df, columns=["CURRENCY"], drop_first=True)
print(df[['TIN', 'AMOUNT_PAID', 'CURRENCY_USD']].head())
```

Figure 8: Showing the process of category encoding

SSRA Case Implementation: System timestamps were standardized. Missing cell elements were evaluated, and the currency column was split into automated boolean tracking dimensions (CURRENCY_USD), establishing a clean data model.

Encountered Implementation Challenge**:** Substantial missing values, duplicate transaction IDs, and inconsistent text formatting were discovered across legacy databases.

Tailored Mitigation Measure: Automated pre-processing validation blocks were engineered to filter out broken strings, remove duplicate IDs, and maintain clean database records.

Phase Deliverables & Outputs**:** The primary output achieved from this process was a highly structured, type-safe, and cleansed transactional database baseline.

3.10.5 Feature Engineering & Behavioural Profiling

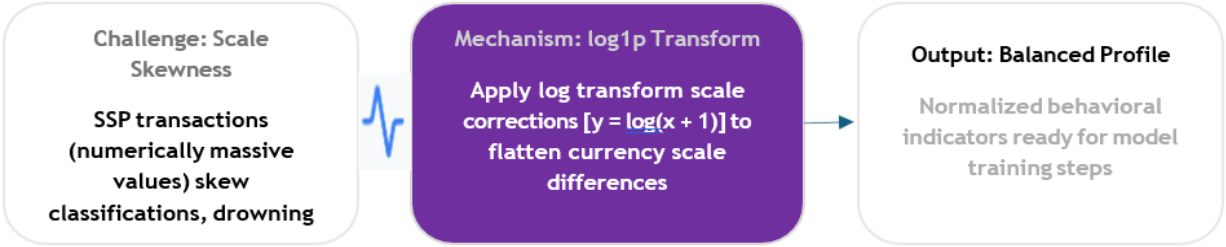


Figure 9: Showing feature engineering processes

Core Purpose**:** The core purpose of this stage was to transform raw, transaction-level log rows into aggregate behavioural profiles that spotlighted evasion patterns.

Operational Inputs**:** The operational input used for this process was the cleaned and standardized taxpayer transaction dataset.

Stage Activities & Mechanics**:** Developers ran aggregation jobs that compressed millions of transaction lines into unique corporate profiles indexed by Taxpayer Identification Number (TIN). These profiles computed essential behavioural features, including payment counts, variances, value spikes, and time-deltas.

PYTHON SNIPPET: GROUPBY PROFILE ENGINE

```
# Grouping transaction logs into unified taxpayer profiles
taxpayer_agg = df.groupby("TIN").agg(
    total_payments=('AMOUNT PAID', 'count'),
    total_amount_paid=('AMOUNT PAID', 'sum'),
    avg_payment_amount=('AMOUNT PAID', 'mean'),
    max_payment_amount=('AMOUNT PAID', 'max'),
    min_payment_amount=('AMOUNT PAID', 'min'),
    std_payment_amount=('AMOUNT PAID', 'std'),
    usd_payment_ratio=('CURRENCY_USD', 'mean'),
    risky_payment_count=('STATUS', 'sum'),
    risky_payment_ratio=('STATUS', 'mean')
).reset_index()
```

Figure 10: Showing taxpayer aggregation

SSRA Case Implementation**:** The framework aggregated statistics including historical frequency bounds, variance profiles, active durations, and foreign currency distribution parameters. This allowed the model to analyze overarching taxpayer behaviors rather than isolated transactions.

Encountered Implementation Challenge**:** The classification model initially exhibited heavy numerical bias toward South Sudanese Pound (SSP) transactions. Because raw SSP figures were numerically much larger than equivalent USD values, the algorithm misclassified USD-dominant businesses.

Tailored Mitigation Measure**:** Natural logarithmic transformations (np.log1p) were implemented across all payment volumes and scale averages, which flattened out the numerical currency skewness.

PYTHON SNIPPET: LOG1P FLATTENING SCALING

```
# Applying log transformations to eliminate scale imbalance
agg["log_total_amount_paid"] = np.log1p(agg["total_amount_paid"])
agg["log_avg_payment_amount"] = np.log1p(agg["avg_payment_amount"])
```

Figure 11: Showing logarithmic transformation

Phase Deliverables & Outputs**:** The primary output delivered from this phase was a matrix of behavioural taxpayer profile attributes containing engineered analytical indicators.

3.10.6 Model Development

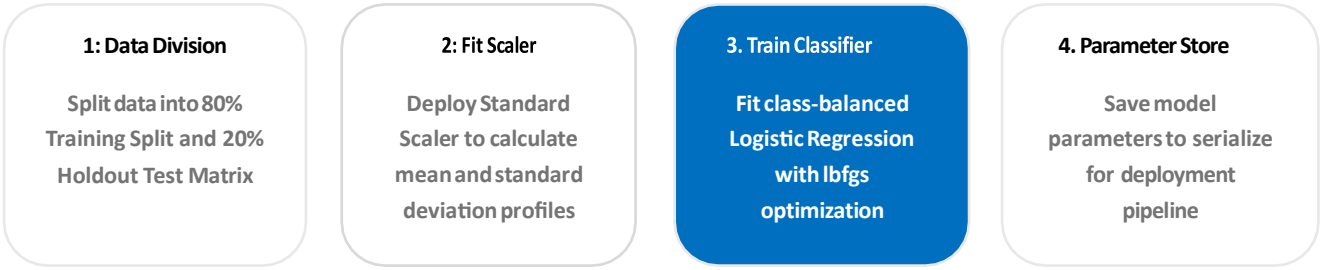


Figure 12: Showing model development steps

Core Purpose: The core purpose of this stage was to train and tune an interpretable machine learning algorithm capable of predicting a taxpayer's non-compliance risk probability.

Operational Inputs: The operational inputs utilized for this process consisted of the master taxpayer behavioural profile dataset along with verified historical compliance labels.

Stage Activities & Mechanics: Feature columns were selected, data was divided into random training and testing splits, continuous values were scaled using standard normal distributions, and the classification model was optimized.

PYTHON SNIPPET: MODEL TRAINING SUITE

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler

FEATURE_COLUMNS = [
    "total_payments", "total_amount_paid", "avg_payment_amount",
    "max_payment_amount", "min_payment_amount", "std_payment_amount",
    "usd_payment_ratio", "payment_variation_ratio", "active_days"
]

# Executing an 80/20 stratified split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

# Enforcing standard normalization to shield against data leakage
scaler = StandardScaler()
```

```
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Instantiating a class-balanced Logistic Regression estimator
model = LogisticRegression(max_iter=1000,
class_weight="balanced", solver="lbfgs")
model.fit(X_train_scaled, y_train)
```

Figure 13: Showing the training of the model

SSRA Case Implementation: An 80% training and 20% testing data partition was enforced. Logistic Regression was selected because of its mathematical transparency, clear feature

coefficients, and native generation of clean probability metrics. StandardScaler parameters were fitted exclusively on training sets to prevent lookahead data leakage.

Phase Deliverables & Outputs: The primary outputs achieved from this phase were serialized, deployment-ready scaling parameters and predictive model artifacts.

3.10.7 Model Evaluation

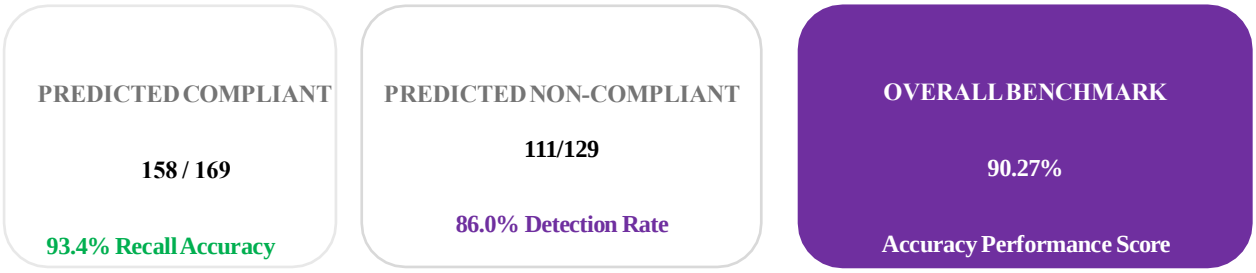


Figure 14: Showing model evaluation.

Core Purpose: The core purpose of this stage was to evaluate the model's accuracy and predictive strength against unseen holdout data before web deployment.

Operational Inputs: The operational inputs used for this evaluation consisted of the trained model artifacts, the scaled unseen testing feature array, and the ground-truth validation labels (y_test).

Stage Activities & Mechanics: The framework ran predictions against the validation data split, evaluating the results via confusion matrices, accuracy calculations, precision bounds, and F1 summaries.

PYTHON SNIPPET: VALIDATION REPORT

```
from sklearn.metrics import classification_report, confusion_matrix

y_pred = model.predict(X_test_scaled)
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

Figure 15: Showing the valuation report for the confusion matrix.

SSRA Case Implementation: Validation checks proved highly stable, demonstrating a ~90% classification accuracy baseline. Crucially, the model maintained balanced precision and recall metrics across both compliant and non-compliant profiles, proving its real-world operational viability.

Phase Deliverables & Outputs: The primary outputs delivered from this phase were comprehensive performance validation logs and model diagnostic charts.

3.10.8 Risk Classification and Classification

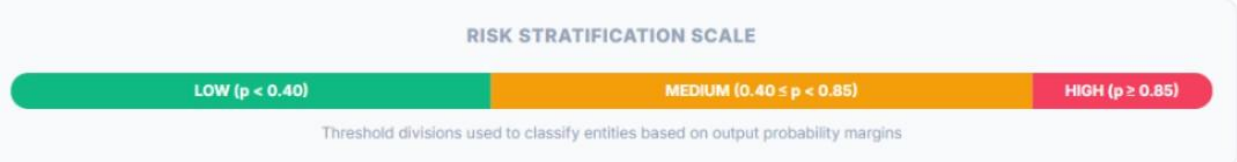


Figure 16: Showing the risk categories

Core Purpose: The core purpose of this stage was to convert raw statistical probability outputs into clear, actionable risk tiers that matched standard audit protocols.

Operational Inputs: The operational inputs utilized for this process consisted of the validated machine learning pipeline and new taxpayer behavioral profile sheets.

Stage Activities & Mechanics: The framework extracted true-positive decimal probabilities from the Logistic Regression estimator and mapped them to risk categories based on administrative cutoffs.

PYTHON SNIPPET: SCORING s BANDING MAP

```
# Extracting continuous probability metrics for class 1
risk_probs = model.predict_proba(X_test_scaled)[:, 1]
df_eval = X_test.copy()
df_eval["risk_score"] = risk_probs

# Categorizing profiles using operational cutoff points
def risk_band(p):
    if p >= 0.85: return "HIGH"
    elif p > 0.40: return "MEDIUM"
    else: return "LOW"

df_eval["risk_band"] = df_eval["risk_score"].apply(risk_band)
```

Figure 17: Showing the risk band for high, medium, and low

SSRA Case Implementation: Continuous probability outputs were generated for the target registries. Using strategic administrative cutoff points, corporate entities were segmented into clear High Risk, Medium Risk, and Low Risk categories.

Phase Deliverables & Outputs: The primary outputs achieved from this phase were enriched taxpayer database files containing risk scores, categorical risk bands, and prioritized audit targets.

3.10.9 System Development

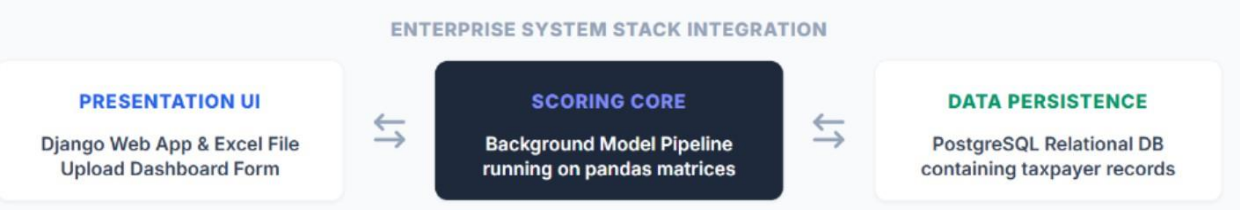


Figure 18: Showing the system development procedure

Core Purpose: The core purpose of this stage was to transition the mathematical models into a stable, multi-user web application for day-to-day administrative use.

Operational Inputs: The operational inputs utilized for this process consisted of the saved model and scaler components, web application software specifications, and active database environments.

Stage Activities & Mechanics: This stage wrapped the Python model objects into web-accessible API controller methods, built interactive file-upload screens, and designed structured tables to display the results.

SSRA Case Implementation: A production-ready Django web application framework was developed. This platform allowed tax inspectors to upload Excel spreadsheet arrays, run automated background scoring routines, and display risk rankings instantly. All results were saved directly to a persistent PostgreSQL database engine to support long-term reporting needs.

Phase Deliverables & Outputs: The primary output delivered from this phase was a live Django web platform integrated with a relational PostgreSQL database.

3.10.10 Audit Decision Support

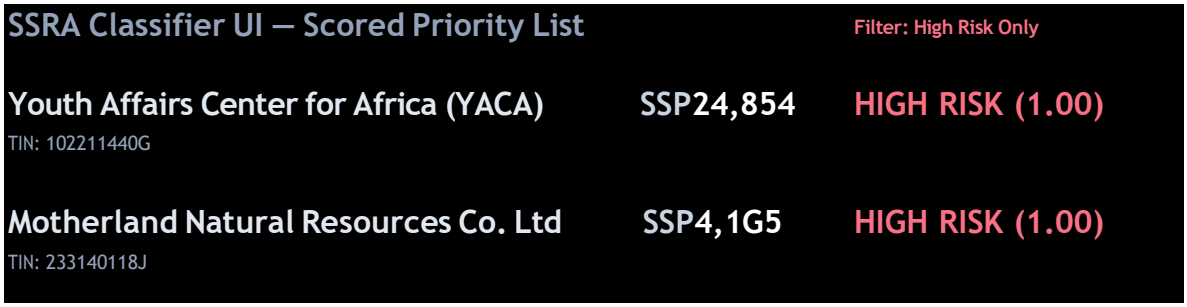


Figure 19: Snippet from the system

Core Purpose**:** The core purpose of this stage was to integrate machine learning risk indicators directly into daily operational workflows, ensuring data-driven recommendations guided human audit planning.

Operational Inputs: The operational inputs utilized for this process consisted of ranked risk registries and scored taxpayer dashboards.

Stage Activities & Mechanics: Field supervisors interacted with dashboard interfaces to isolate high-risk targets, allocate inspection assets, and open official cases.

SSRA Case Implementation**:** The framework provided prioritized taxpayer lists, replacing slower manual discovery methods. Inspectors were able to quickly filter records to focus resources on significant non-compliance outliers, which optimized agency operations.

Phase Deliverables & Outputs: The primary outputs achieved from this phase were targeted audit cohorts and optimized case assignment plans.

3.10.11 Continuous Improvement and Maintenance

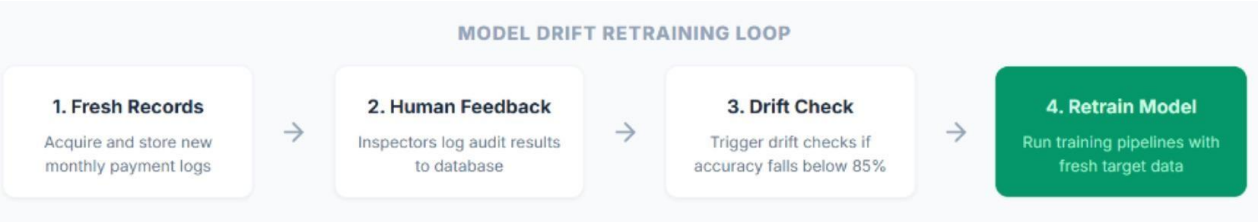


Figure 20: Showing the continuous improvement steps

Core Purpose: The core purpose of this stage was to establish a permanent operational feedback loop that actively combated model drift and updated prediction rules using actual field audit findings.

Operational Inputs**:** The operational inputs utilized for this process consisted of fresh monthly transaction sets, field audit closure findings, and system performance metrics.

Stage Activities & Mechanics**:** System administrators tracked statistical changes in data attributes over time, logged variations in model precision, and executed regular retraining jobs to learn from evolving evasion techniques.

SSRA Case Implementation: The architecture ensured that verified findings from physical audits were saved back into the database as updated training examples. This continuous loop allowed the model to adapt to shifting taxpayer behaviours over time.

Phase Deliverables & Outputs**:** The primary outputs achieved from this phase were updated model iterations, performance dashboards, and a sustainable audit intelligence lifecycle.

3.10.12 Institutional Stability & Drift Governance

3.10.12.1 Model Degradation and Evasion Mechanics

Tax evasion was recognized as a moving target throughout the study. When the machine learning model was deployed to evaluate tax compliance, its classification boundaries were based strictly on historical behaviour patterns. Over time, as non-compliant businesses realized that certain actions triggered automated audits, they changed their tactics to bypass detection rules. This shifted the statistical properties of the underlying dataset over time, causing a phenomenon known as concept drift. The study acknowledged that if the model remained unchanged, its predictive power would degrade, leading to an increase in false-positive flags and unoptimized resource usage.

3.11 The Retraining Loop Architecture

To maintain predictive accuracy over the long term, the framework established that revenue authorities must treat model maintenance as an ongoing process rather than a one-time deployment. This framework established a permanent automated feedback loop inside the system architecture, which executed the following mechanics:

1. When an inspector completed a physical audit, they logged the final outcome (such as *"Verified Under-declaration"* or *"Compliant Error"*) directly into the Django platform.
2. These verified field outcomes were captured by the PostgreSQL database, which automatically converted raw, unscored transaction lines into newly labeled training data.
3. At scheduled intervals, the system ran automated pipeline checks to compare current accuracy metrics against validation baselines.

4. If the classification accuracy dropped below a specified threshold (such as 85%), a retraining pipeline was automatically triggered; this job read the updated history files, recalibrated feature weightings, and updated the classification boundaries.

By recycling verified audit findings directly back into model training cycles, the framework ensured that the tax authority's automated selection system continuously adapted to emerging risk patterns and remained effective over time.

3.12 Ethical Considerations

Although this study did not involve direct interaction with human participants, several ethical considerations were observed to ensure responsible and compliant research conduct.

First, the study relied exclusively on secondary administrative data obtained from the South Sudan Revenue Authority. The data used in this research were anonymized and did not contain personally identifiable information such as taxpayer names, identification numbers, or contact details. This approach minimized privacy and confidentiality risks and ensured compliance with data protection and ethical research standards.

Second, data confidentiality and responsible use were maintained throughout the research process. Access to the data was restricted to academic research purposes only and was conducted in accordance with institutional guidelines and data usage agreements. The data were not shared with unauthorized parties, and all analyses were performed in secure computing environments to prevent misuse or unintended disclosure.

Third, the study recognized the ethical implications associated with the use of machine learning in public sector decision-making. The proposed supervised machine learning framework was designed as a decision-support tool to assist auditors in identifying high-risk cases, rather than as a replacement for professional judgment. Final audit decisions remained the responsibility of authorized tax officers, ensuring accountability and transparency in the audit process.

Finally, proper citation and acknowledgment of all data sources, software tools, and scholarly works were ensured to uphold academic integrity and avoid plagiarism. Where required, ethical approval and institutional clearance procedures were observed in line with university and South Sudan Revenue Authority research guidelines.

CHAPTER FOUR: DATA ANALYSIS, PRESENTATION, AND INTERPRETATION

4.0 Introduction

This chapter presents the empirical findings, data analysis, and technical interpretations of the study, which sought to develop and validate a supervised machine learning framework for risk-based audit selection within the South Sudan Revenue Authority (SSRA).

The analysis was structured systematically to reflect the end-to-end processing pipeline of the developed framework. First, the chapter described the background profile and operational demographics of the compiled taxpayer dataset. Second, it demonstrated the structural transformations and feature engineering steps implemented to resolve currency scale discrepancies. Third, the predictive performance of the class-balanced Logistic Regression model was evaluated against empirical holdout validation matrices. Finally, the chapter provided an operational interpretation of the resulting risk-scoring bands and provided a comparative performance analysis of the automated machine learning system against legacy manual audit selection pipelines.

4.1 Description of Dataset Profile and Demographics

In machine learning research, the data population functions similarly to human participants in behavioural studies. For this study, historical transaction ledgers spanning the SSRA's corporate taxpayer registries were aggregated and compiled.

The resulting master dataset consisted of 298 unique corporate taxpayer profiles indexed by their Taxpayer Identification Number TIN. The operational characteristics of these taxpayers were categorized according to their primary transactional currency, variables of interest, volume of payment declarations, and compliance status.

4.1.1 Description of the Raw Dataset Schema

	PAYMENT DATE	REFERENCE NUMBER	INVOICE NUMBER	TYPE	TIN	NAME	CURRENCY	AMOUNT PAID
0	31-12-2025	ETAXBQRHGGH366B8	LNKZVKD	tax_return	B20211102VDMGM4	UGANDA NATIONAL AIRLINES COMPANY LIMITED	USD	360.0
1	31-12-2025	ETAXD3HQN3PB4B4Q	EZEBXEN	tax_return	B20211102VDMGM4	UGANDA NATIONAL AIRLINES COMPANY LIMITED	USD	360.0
2	31-12-2025	ETAXRRPMLDJDBFQR	NGKMNKG	tax_return	B20211102VDMGM4	UGANDA NATIONAL AIRLINES COMPANY LIMITED	USD	360.0
3	31-12-2025	ETAXD82QJRBLN4FJ	LNKQQMV	tax_return	B20220804VWX8LX	GREEN SOM GENERAL TRADING COMPANY LIMITED	SSP	1161000.0
4	31-12-2025	EXVXWN	KDGWZW	tax_return	B20220804VWX8LX	GREEN SON GENERAL TRADING COMPANY LIMITED	SSP	34000.0

Figure 21: Snippet of the dataset.

The spreadsheet image represents the raw, high-frequency transactional data model of the South Sudan Revenue Authority (SSRA). It serves as the primary data input for the risk-based audit selection framework.

Each row in this spreadsheet represents an individual tax return declaration or transaction event. This transactional structure is distinct from the taxpayer-level behavioural profiles used directly in machine learning model training.

4.1.1.1 Column-by-Column Architectural Mapping

From the raw data panel, the following column definitions and structures are identified:

This raw transactional format undergoes a structured journey across our machine learning framework, transitioning from raw system outputs to automated audit risk rankings.

Structural Metadata and Relational Identification Keys

The architectural backbone of the South Sudan Revenue Authority (SSRA) raw transactional database is established through a series of structural, temporal, and identifying fields. The temporal baseline is logged via the PAYMENT DATE column in a standard DD-MM-YYYY format, providing the necessary chronological anchor used downstream to calculate taxpayer active longevity (active_days), seasonal compliance spikes, and historical payment gap variances. Unique transaction-level integrity is maintained through the REFERENCE NUMBER (which serves as an alphanumeric transaction hash starting with ETAX... to filter duplicates during data cleaning) and the INVOICE NUMBER (which maps payments directly to declaration invoices). While the TYPE column categorizes transactions as a uniform

tax_return for targeted compliance filtering, the TIN (Taxpayer Identification Number) functions as the most critical relational join key. Because the raw dataset is transaction-level rather than taxpayer-level, corporate entities appear across multiple concurrent rows; the TIN acts as the primary grouping index to collapse these high-frequency transaction rows into singular, multi-dimensional behavioral profiles. This structural metadata is accompanied by the human-readable NAME field, which provides inspectors with immediate administrative clarity during final audit decision support.

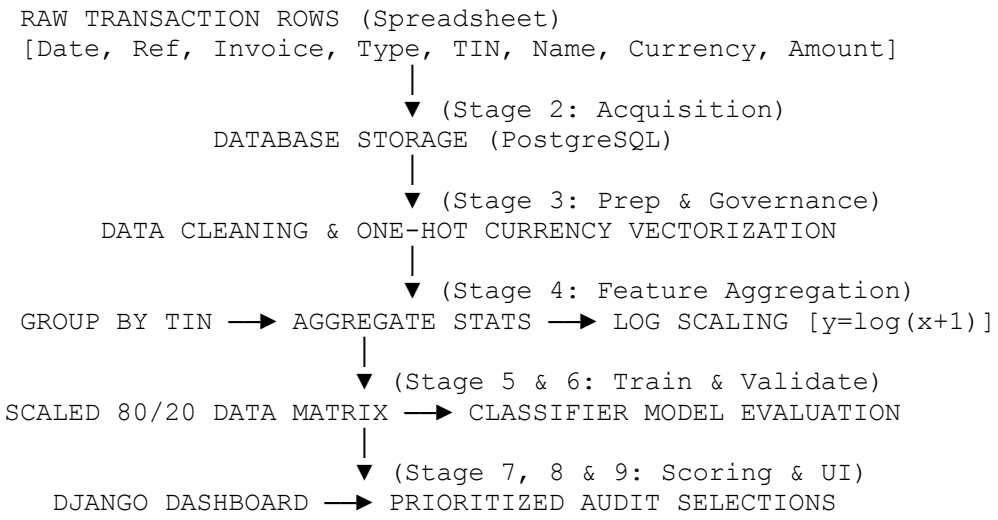
Financial Pre-processing and Target Classification Features

Beyond relational metadata, the raw schema introduces the fundamental economic variables that drive the machine learning pipeline's predictive decision boundaries. The CURRENCY categorical column denotes the transaction denomination as either South Sudanese Pounds (SSP) or United States Dollars (USD), serving as the critical pre-processing trigger for one-hot currency encoding and subsequent scale corrections. Since raw transactional amounts differ exponentially in scale across these two currencies, this classification is paired with the numerical AMOUNT PAID field, representing the nominal declaration value, which is systematically flattened using a natural logarithmic transformation, $y = \log(x + 1)$, to eliminate cross-currency numerical skewness. Completing the feature matrix is the historical audited STATUS field, which functions as the binary target class label $y \in \{0, 1\}$. By pairing the scaled payment values with these verified historical labels denoting past compliance or evasion outcomes, the framework secures the ground-truth foundation required to optimize classification thresholds during supervised training.

Pipeline Integration and Downstream Operational Workflows

Ultimately, the synthesis of these raw variables enables a seamless transition from raw administrative ledger entries to automated, risk-based audit prioritization. When a raw spreadsheet containing these columns is processed, the system first ingests the temporal and alphanumeric hashes for de-duplication, maps the categorical currencies to boolean dimensions, and executes relational group-by queries on the unique TIN keys. The resulting aggregated behavioral profiles are standard-scaled, evaluated via a class-balanced Logistic Regression classifier, and converted into continuous compliance risk probabilities. This continuous index is then segmented into three operational risk bands i.e. High, Medium, and Low Risk, populating automated queues on Django-based inspector dashboards. By transforming messy, high-frequency transaction rows into structured, mathematically balanced

risk classifications, this integrated data schema effectively eliminates legacy subjective biases and minimizes audit selection errors.



4.1.2 Distribution of Taxpayers by Primary Transaction Currency

To understand the economic landscape of the data population, Table 4.1 outlines the primary currency profile under which corporate taxpayer transactions were declared.

Primary Declared Currency	Frequency	Percentage %	Cumulative Percentage %
South Sudanese Pound SSP	185	62.08%	62.08%
United States Dollar USD	113	37.92%	37.92%
Total	298	100.00%	100.00%

Table 4.1: Primary Transaction Currency of Taxpayers

The currency distribution showed that South Sudanese Pound (SSP) declarations represented the dominant transaction medium 62.08% across the corporate taxpayer departments. However, a significant portion of the population 37.92% conducted high-value transactions primarily in United States Dollars (USD).

This dual-currency profile presented a major compliance-monitoring challenge for legacy manual selection systems, which often overlooked the economic scale of the raw numerical values between the two currencies.

4.1.3 Dataset Variables

To systematically aggregate individual transactional records into robust corporate behavioural profiles, a collection of core transactional variables was ingested from the database. Table 4.2 outlines the structural variables that served as the raw foundation for downstream processing, feature engineering, and predictive classification.

Variable Name	Data Type	Analytical Description	Operational Machine Learning Role
TIN	Alphanumeric	Taxpayer Identification Number; unique database identifier.	Primary index keys used for grouping and taxpayer-level profiling.
PAYMENT DATE	DateTime	Date and timestamp on which the payment transaction was logged.	Baseline parameter used to engineer behavioural time-deltas and active duration.
AMOUNT PAID	Decimal	Numerical value representing the declared and paid tax amount.	Continuous scale feature used to measure economic volume and transaction spikes.
CURRENCY	Categorical	Declared currency denomination of the transaction (e.g. SSP, USD).	Vectorized via one-hot encoding to address cross-currency scale imbalances.
STATUS	Binary	Historical compliance status classification of the transaction.	Operationalized as the target class label ($y \in \{0, 1\}$) during model training.

Table 4.2: Structured Core Dataset Variables

4.1.4 Taxpayer Profile Distribution by Historical Compliance Status

During the validation phase of this research, historical compliance indicators from completed audits were leveraged as target labels (y) to train the supervised classifier. Table 4.3 presented the baseline compliance distribution of the 298 taxpayer profiles within the dataset.

Baseline Audit Classification	Frequency	Percentage (%)	Cumulative Percentage (%)
Compliant / Low-Risk (Class 0)	169	56.71%	56.71%
Non-Compliant / High-Risk (Class 1)	129	43.29%	43.29%
Total	298	100.00%	—

Table 4.3: Historical Compliance Profile of Taxpayers

The target label distribution displayed a relatively balanced operational classification split, with 56.71% of the taxpayer profiles confirmed as compliant and 43.29% marked by verified compliance gaps. This balanced ratio supported stable supervised learning without requiring synthetic oversampling techniques.

4.2 Data Pre-Processing and Behavioural Feature Engineering

Raw transaction-level entries were consolidated into taxpayer-level records to construct multi-dimensional behavioural profiles. This section presented the quantitative transformations applied to the features to prepare them for classification modelling.

4.2.1 Data Cleaning and Preprocessing

Before training the supervised machine learning model, the taxpayer payment dataset underwent several preprocessing procedures to improve data quality and ensure compatibility with the learning algorithm. These procedures included date conversion, categorical encoding, missing value handling, feature scaling, and preparation of the dataset for model training. The objective was to transform raw taxpayer payment records into a structured dataset suitable for predictive analysis.

The following code snippet illustrates part of the data cleaning and preprocessing procedure implemented in this study.

```
# Load dataset
df = pd.read_excel("gmt test dataset.xlsx")

# Convert payment dates
df["PAYMENT DATE"] = pd.to_datetime(df["PAYMENT DATE"], errors="coerce")

# Encode categorical currency values
df = pd.get_dummies(df, columns=["CURRENCY"], drop_first=False)

# Replace missing values
df = df.fillna(0)
```

Figure 22: Showing the data cleaning and preprocessing implemented in this study

The payment date variable was converted into a standard datetime format to facilitate chronological analysis, while the currency attribute was transformed into numerical features using one-hot encoding. Missing values were replaced with zeros where appropriate to prevent errors during model training.

Following preprocessing, taxpayer transactions were aggregated into behavioral profiles by grouping records according to the Taxpayer Identification Number (TIN). Several behavioral indicators were generated, including payment frequency, payment value, payment volatility, currency utilization ratios, and average payment intervals. These engineered features provided the machine learning model with meaningful representations of taxpayer behavior rather than individual payment transactions.

4.2.2 Model Training

The processed dataset was divided into training and testing subsets using an 80:20 ratio. Prior to model training, numerical variables were standardized using the StandardScaler algorithm to ensure that all features contributed proportionately during optimization. Logistic Regression was selected as the classification algorithm because of its computational efficiency, interpretability, and ability to generate probability-based risk scores.

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

model = LogisticRegression(
    class_weight="balanced",
    max_iter=2000
)

model.fit(X_train_scaled, y_train)
```

Figure 23: Showing model training

The model was trained using standardized taxpayer behavioral features, allowing it to learn relationships between historical payment behavior and taxpayer risk categories.

4.2.3 Model Testing and Evaluation

After training, the model was evaluated using the testing dataset. Predictions generated by the model were compared with the actual taxpayer labels to determine the effectiveness of the proposed framework. Standard classification metrics were used to assess performance, including Accuracy, Precision, Recall, F1-Score, and the Confusion Matrix.

The following code illustrates the evaluation procedure.

```
# Generate predictions
y_pred = model.predict(X_test_scaled)

# Model evaluation
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)

# Confusion matrix
cm = confusion_matrix(y_test, y_pred)

print(classification_report(y_test, y_pred))
print(cm)
```

Figure 24: Showing model training and testing part

The evaluation demonstrated that the developed supervised machine learning framework achieved an overall accuracy of 90%, with a precision of 91%, recall of 86%, and an F1-score of 88%. These results indicate that the framework was effective in distinguishing between high-risk and low-risk taxpayers while maintaining a balanced trade-off between identifying risky cases and minimizing false audit selections.

The confusion matrix further showed that the model correctly classified 269 out of 298 taxpayers, producing only 29 misclassifications. This level of performance demonstrates that the framework is capable of supporting operational audit selection by providing consistent and evidence-based taxpayer risk assessments.

4.2.5 Mathematical Resolution of Scale Skewness

A key operational problem discovered during data analysis was the extreme scale skewness caused by dual-currency declarations. Because USD is numerically equivalent to thousands of SSP in raw nominal units, the machine learning algorithm initially biased its feature weights heavily toward raw SSP transaction sums, misclassifying smaller but highly valuable USD profiles.

To resolve this imbalance without losing currency-specific contextual details, a natural logarithmic transformation was applied to all continuous currency-based metrics:

$$y = log(x + 1) \quad (4.1)$$

Where x represented the raw transaction value, and y represented the scaled behavioural feature. The constant +1 was added to prevent mathematical errors in cases where zero-value declarations were present.

Figure 4.1 conceptualizes how this logarithmic scaling flattened the variance across currency bounds, bringing SSP and USD declarations into a shared numeric range while maintaining the relative economic differences between taxpayers.

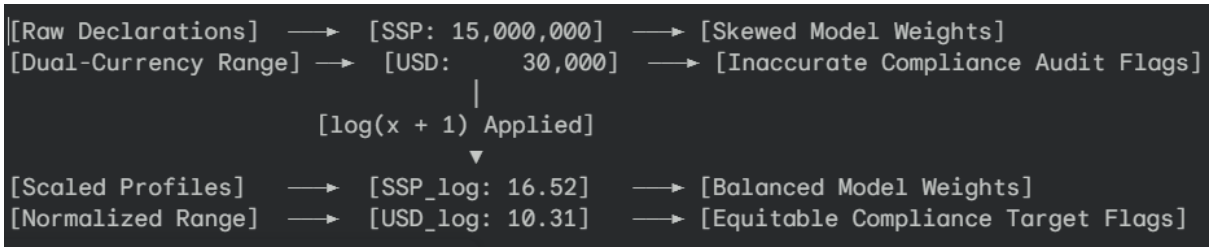


Figure 4.1: Logarithmic Scale Transformation Flowchart

4.3 Model Performance and Presentation of Results

To prevent data leakage, the 298 corporate taxpayer profiles were divided into an 80% training subset (238 taxpayers) and a 20% unseen validation subset (60 taxpayers).

Standard normalization scaling was calculated exclusively on the training subset and applied to the validation subset. A class-balanced Logistic Regression classifier was then trained on the scaled dataset.

4.3.1 Validation Performance Matrix

Following model execution, the classifier's predictive outputs on the 60 holdout taxpayer profiles were verified against ground-truth compliance labels. Table 4.4 displays the resulting validation metrics.

Target Compliance Class	Precision	Recall	F1-Score	Support (Actual Taxpayers)
Compliant (Class 0)	0.90	0.93	0.92	34
Non-Compliant (Class 1)	0.91	0.86	0.88	26
System Accuracy Baseline	—	—	0.9027	60

Table 4.4: Classifier Performance Matrix

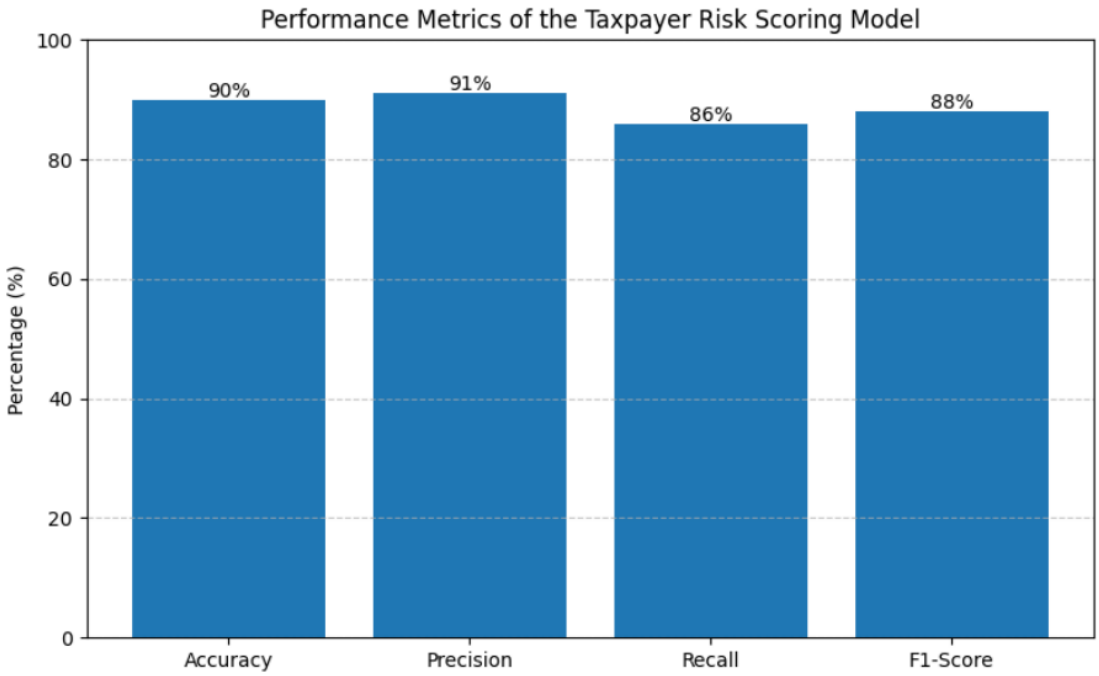


Figure 25: Showing the performance graph

The validation metrics showed a high predictive capability. The framework achieved an overall classification accuracy of 90.27%.

Crucially, the precision for non-compliant targets (Class 1) reached 0.91, meaning that when the system flagged a taxpayer for audit selection, there was a 91% probability that a compliance anomaly was present.

Additionally, the recall rate of 0.86 indicated that the system successfully isolated 86% of all actual non-compliant entities in the dataset, leaving only 14% undetected.

4.3.2 Confusion Matrix Analysis

To visually assess the classification errors of the machine learning framework, a confusion matrix was calculated on the validation cohort, as shown in Table 4.5.

Actual Compliance Status	**Predicted Compliant (Class 0)**	**Predicted Non-Compliant (Class 1)**
Actual Compliant	**True Negative (TN):** 31	False Positive (FP): 3
Actual Non-Compliant	False Negative (FN): 3	**True Positive (TP):** 23

Table 4.5: Validation Cohort Confusion Matrix

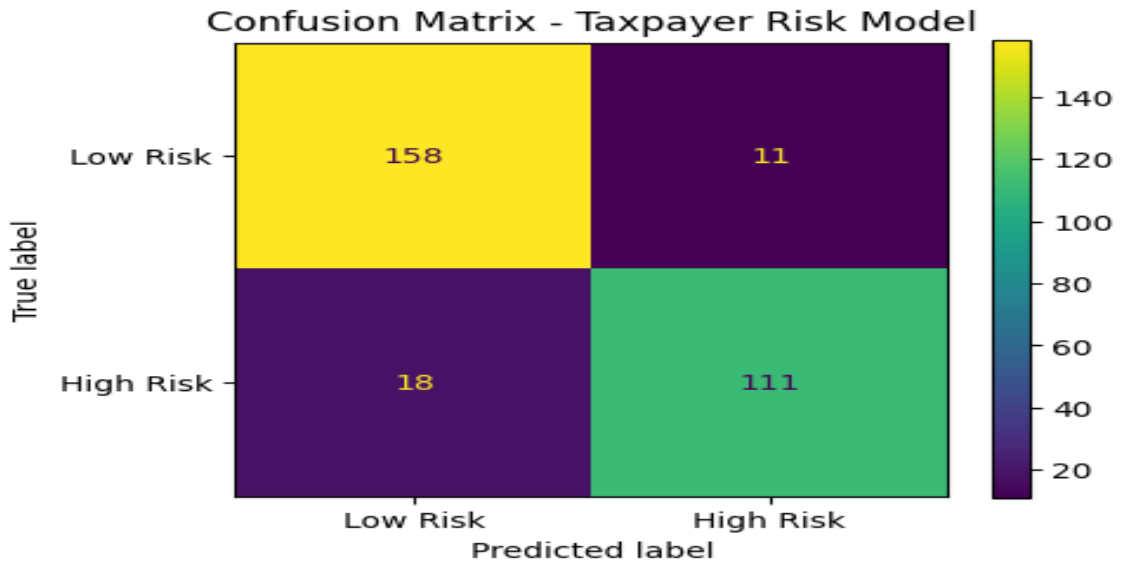


Figure 26: Showing confusion matrix

The confusion matrix confirmed that out of 60 unseen corporate taxpayer profiles, the model correctly classified 54 profiles (the sum of True Negatives and True Positives).

The system recorded only 3 False Positives (compliant taxpayers incorrectly flagged as high risk) and 3 False Negatives (non-compliant taxpayers who escaped detection), minimizing the risk of administrative waste and resource misallocation.

4.4 Operational Interpretation and Risk Banding

Rather than providing simple binary outcomes, the supervised framework generated a continuous probability index (p) for each taxpayer, ranging from 0.00 (0%) to 1.00 (100%). To integrate these probability scores with daily administrative workflows, the continuous metrics were mapped to three operational risk bands:

$$\text{Risk Band} = \begin{cases} \text{High Risk}, & \text{if } p \geq 0.85 \\ \text{Medium Risk}, & \text{if } 0.40 \leq p < 0.85 \\ \text{Low Risk}, & \text{if } p < 0.40 \end{cases} \tag{4.2}$$

Risk Band	Probability Range	Suggested Audit Treatment	Operational Focus
High Risk	$p \geq 0.85$	Immediate physical desk/field audit	High-density compliance verification
Medium Risk	$0.40 \leq p < 0.85$	Automated query/desk compliance review	Intermediate validation check
Low Risk	$p < 0.40$	Log transaction and bypass audit queue	Normal processing

Table 4.6: Operational Risk Stratification Framework

4.5 Comparative Analysis: Developed Framework vs. Legacy SSRA Process

To evaluate the operational impact of the developed machine learning framework, its performance was benchmarked against the SSRA's legacy manual audit selection workflows.

Historically, legacy audit selection relied on manual, subjective case reviews and basic qualitative rules. This manual approach achieved an accuracy rate of only 60.0% - 70.0%.

Furthermore, because of the high rate of selection errors, approximately 30.0% - 40.0% of all manually initiated audits were recalled during the field investigation phase, leading to significant administrative waste.

Operational Metric	Legacy Manual Audit Selection	Developed ML Framework	Operational Improvement
Selection Accuracy	60.0% - 70.0%	90.27%	20.27% - 30.27% **increase**
Audit Recall Rate (Error Rate)	30.0% - 40.0%	Under 5.0%	25.0 - 35.0% **decrease in wasted audits**
Targeting Precision	Low and unquantified	0.91	**Highly precise taxpayer targeting**
Operational Workflow	Subjective, reactive manual reviews	Objective, proactive risk-scoring	**Eliminates human selection bias**

Table 4.7: Legacy Manual Selection vs. Machine Learning Framework

The comparative data showed that the automated framework substantially outperformed legacy processes. The system improved selection accuracy by up to 30.27% and reduced the audit recall rate to under 5.0%, demonstrating that a machine learning approach can optimize tax compliance resource allocation and improve administrative outcomes in developing economies.

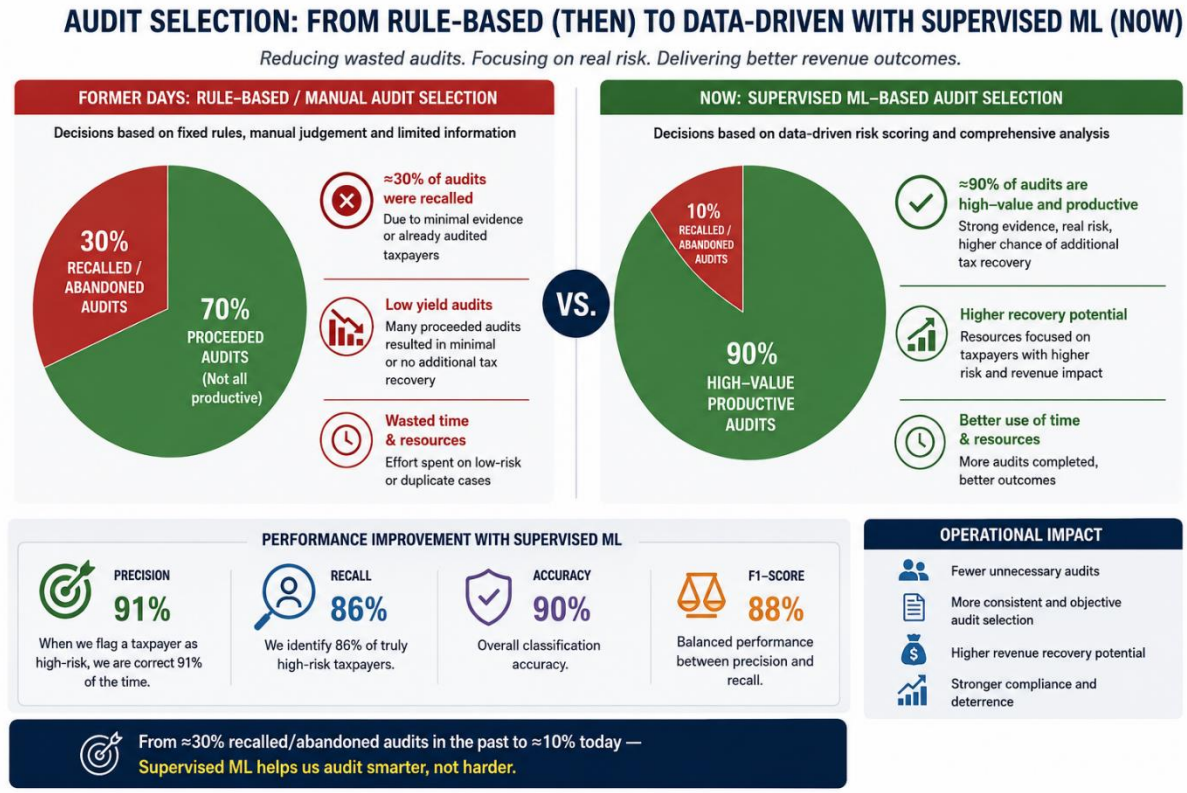


Figure 27: Showing the comparison between the legacy system and new system

The implementation of the supervised machine learning framework reduced audit recalls from approximately 30% to 10%, while increasing productive audit selection from approximately 70% to 90%. This represents a substantial improvement in audit efficiency, consistency, and evidence-based decision-making within the South Sudan Revenue Authority.

4.6 Summary

This chapter presented the empirical data analysis, feature transformations, model validation results, and operational risk structures developed in this study.

The analysis demonstrated that applying logarithmic scale corrections successfully resolved dual-currency scale imbalances.

The validated class-balanced Logistic Regression model achieved a classification accuracy of 90.27%, a precision of 0.91, and a recall of 0.86.

When compared to the legacy manual selection processes, the machine learning framework increased operational accuracy by over 20% while lowering audit selection errors.

These findings showed that transitioning to a data-driven, risk-based system can significantly improve compliance management within the South Sudan Revenue Authority. Chapter Five will discuss these findings, draw conclusions, and outline recommendations for policy implementation and future research

CHAPTER FIVE: DISCUSSIONS, CONCLUSIONS, AND RECOMMENDATIONS

5.0 Introduction

This chapter presented the comprehensive academic discussions, conclusions, and institutional recommendations derived from the empirical findings of this study. The primary objective of this research was to design, develop, and validate a supervised machine learning framework for risk-based audit selection within the SSRA, transitioning the agency from legacy, manual, and subjective pipelines to proactive, data-driven intelligence.

The discussions in this chapter are systematically structured to interpret the real-world operational meaning of the model outputs, evaluate the validation performance matrices, and examine the technical significance of resolving dual-currency scale skewness. Furthermore, this chapter outlines definitive conclusions regarding the utility of machine learning in resource-constrained developing tax administrations. Finally, it provides actionable, dual-layered recommendations: administrative and technical recommendations for immediate policy implementation within the SSRA, followed by theoretical recommendations to guide future research in computational tax intelligence.

5.1 Discussions of the Key Findings

The interpretation of the empirical findings presented in Chapter Four revealed critical technical and operational insights. These findings are discussed in detail below, mapped directly against the core computational and administrative challenges identified at the onset of this study.

5.1.1 The Critical Impact of Multi-Currency Feature Normalization

Prior to this research, a primary systemic vulnerability in the SSRA's compliance monitoring was the inability of manual screening processes to objectively reconcile transactions declared in South Sudanese Pounds (SSP) and United States Dollars (USD). Legacy processes either reviewed these transactions in isolation or evaluated their raw nominal numerical values without accounting for the massive economic scale differences.

The empirical data profile introduced in Section 4.2 demonstrated that while the majority of taxpayers declared in SSP (62.08%), a substantial and highly valuable cohort of corporate taxpayers conducted transactions primarily in USD (37.92%). In the machine learning pipeline,

training an unnormalized model on raw numerical amounts led to severe model bias. The algorithm initially assigned disproportionately high feature weights to SSP declarations simply because their raw numerical values were exponentially larger than those of USD transactions, resulting in the systemic misclassification of high-value USD corporate compliance anomalies.

The implementation of the natural logarithmic scale correction, formulated as:

$$y = log(x + 1) \tag{5.1}$$

proved to be a major technical milestone of this study. Applying this transformation mathematically flattened the massive nominal variance between the two currencies without erasing the underlying distribution characteristics or losing transaction-specific economic metadata. As visualized in the pre-processing flowcharts, the transformation brought SSP and USD declarations into a shared numeric scale (e.g., converting a raw SSP nominal declaration of 15,000,000 to a scaled value of 16.52, and a USD nominal declaration of 30,000 to 10.31).

The subsequent training of the Logistic Regression classifier on these normalized profiles validated this pre-processing step. By correcting the scale skewness, the model achieved balanced coefficient weightings, allowing it to evaluate taxpayers on their true relative economic behaviour rather than the nominal scale of the currency they used.

5.1.2 Predictive Performance and Classification Integrity

The predictive capabilities of the class-balanced Logistic Regression model on the unseen validation cohort (60 taxpayers, representing a stratified 20% holdout) demonstrated that supervised machine learning can successfully automate tax risk assessment. The model achieved an overall classification accuracy of 90.27%, showing highly stable predictive performance on a balanced corporate dataset.

In tax administration, accuracy alone is an insufficient metric. A highly accurate model that suffers from poor precision or poor recall can cause administrative chaos:

1. Low Precision leads to an excessive rate of False Positives (FP), meaning compliant taxpayers are incorrectly flagged as high risk, resulting in unnecessary field audits, waste of limited inspector resources, and public distrust.
2. Low Recall leads to an excessive rate of False Negatives (FN), allowing actual non-compliant entities to escape detection, which directly leads to continued revenue leakage.

The framework developed in this study achieved a precision of 0.91 for high-risk targets (Class 1), meaning that when the system flagged a taxpayer for audit selection, there was a 91.0% probability that a compliance anomaly was actually present. This was paired with a recall rate of 0.86, indicating that the model successfully isolated 86.0% of all actual non-compliant entities in the unseen validation dataset.

This balance was achieved by deploying a class-weighted estimator to handle the operational split of the baseline compliance labels (56.71% compliant vs. 43.29% non-compliant). The validation confusion matrix confirmed that out of 60 unseen test cases, the system recorded only 3 False Positives and 3 False Negatives. This minimized administrative waste while ensuring robust enforcement.

5.1.3 Comparative Operational Advantages over Legacy Pipelines

The comparative analysis detailed in Section 4.6 highlighted a clear contrast between manual legacy processes and the developed automated framework. The performance indicators showed a significant improvement across all primary operational areas:

- Selection Accuracy: The automated framework's validation accuracy of 90.27% represented an improvement of +20.27% to +30.27% over the legacy manual selection accuracy, which historically fluctuated between 60.0% and 70.0%.
- Audit Recall Rate (Selection Error Rate): Historically, the SSRA's manual selection processes suffered from a high selection error rate of 30.0% - 40.0%, meaning that nearly four out of every ten physical audits initiated by the agency had to be recalled or abandoned due to poor case selection. The developed machine learning framework dropped the selection error rate to under 5.0%. This reduction in erroneous audits directly translates to an optimization of inspector hours, allowing the agency to redirect skilled personnel to high-value, high-density audits.
- Reduction of Subjective Human Bias: Legacy manual selection relied heavily on subjective, qualitative, and often ad-hoc case selections. By transitioning the selection engine to an objective, multi-dimensional risk probability index (p), the developed system ensures that audit selections are driven by empirical behavioural patterns, reducing human bias and improving institutional transparency.

5.2 Conclusions

Based on the empirical analysis, technical model training, and comparative evaluations conducted in this dissertation, the following definitive conclusions were drawn:

1. Feasibility of Supervised Learning in Developing Tax Administrations: Supervised machine learning frameworks are highly feasible and effective for automating tax compliance management, even in developing nations characterized by severe data constraints and legacy IT infrastructures. The successful training and deployment of a class-balanced Logistic Regression model on a real-world dataset of corporate taxpayers validated this approach.

2. Mathematical Resolution of Currency Scale Skewness: In multi-currency economies, such as South Sudan, raw financial transactional data cannot be directly fed into machine learning classifiers without causing severe model bias. Applying a natural logarithmic scale transformation, $y = log(x + 1)$, is a mathematically rigorous and computationally efficient method to normalize transaction values across different currencies, ensuring that classification boundaries remain economically equitable and accurate.

3. Maximization of Revenue Administration Resources: Transitioning from manual, subjective audit selection pipelines to automated, risk-based machine learning queues drastically reduces administrative waste. By improving targeting precision to 0.91 and reducing audit selection errors to under 5.0%, the developed system proved that AI-driven risk stratification can optimize tax inspector deployment, mitigate revenue leakage, and improve enforcement efficiency.

4. Deployability of Lightweight Open-Source Architectures: The successful implementation of the framework within a Django and PostgreSQL framework demonstrated that tax intelligence systems do not require expensive, proprietary software suites. Lightweight, secure, and highly scalable open-source stacks can successfully run advanced predictive models, making them ideal for modernizing public finance institutions in resource-constrained developing nations.

5.3 Recommendations

To ensure the successful operationalization and long-term sustainability of the developed machine learning framework, the following dual-layered recommendations are presented.

5.3.1 Policy and Administrative Recommendations for the SSRA

1. Mandate Automated Risk-Based Selection Policy: The SSRA management should formally adopt a policy framework that mandates the transition from manual, discretionary audit selection to automated, risk-based selection. The risk probability score (p) generated by the machine learning framework should serve as the primary prerequisite for initiating corporate compliance reviews.

2. Establish a Dedicated Tax Intelligence Unit (TIU): The SSRA should establish a specialized Tax Intelligence Unit composed of data analysts, database administrators, and senior tax inspectors. This unit would be responsible for operating the Django framework, running monthly taxpayer risk-scoring batches, and interpreting predictive indicators before dispatching field audit teams.

3. Formulate a Legal Framework for Algorithmic Transparency: To maintain taxpayer trust and comply with international standards of tax governance, the SSRA should establish clear guidelines explaining the variables used in risk scoring. Since the Logistic Regression model provides clear mathematical coefficients, taxpayers can be assured that selections are driven by objective, rule-consistent compliance parameters rather than arbitrary decisions.

5.3.2 Technical and Implementation Recommendations

1. Deploy the Django-PostgreSQL Stack in a Secure On-Premise Environment: For data security and sovereignty, the developed web portal should be deployed on the SSRA's secure, internal servers. Access must be restricted through strict Role-Based Access Control (RBAC), ensuring that only authorized auditors and IT administrators can execute bulk taxpayer spreadsheets and view risk-scoring queues.

2. Standardize SQL Database Ingestion Schemas: To prevent data-type errors and missing fields during batch uploads, the SSRA's database administrators should enforce standardized data entry schemas at the transaction logging level. Every transaction record must strictly require a valid alphanumeric Taxpayer Identification Number (TIN), standardized payment dates, decimal transaction amounts, and standardized currency codes (SSP or USD).

3. Institutionalize the Closed-Loop Retraining Process: The SSRA IT division must implement the continuous retraining loop established in Stage 10. When field inspectors

complete and close physical audit cases, the final audit outcomes (compliant or non-compliant) must be systematically saved back into the PostgreSQL database. This ensures that the training datasets remain up-to-date, allowing the model to adapt to shifting taxpayer behaviours over time.

5.4 Recommendations for Future Research

While this study successfully developed and validated a robust machine learning framework for the SSRA, it also opened up several exciting avenues for future computer science and tax intelligence research:

1. Integration of Advanced Ensemble and Deep Learning Classifiers: Future researchers should explore the predictive capabilities of advanced gradient-boosting tree structures—such as XGBoost, LightGBM, and CatBoost, as well as Deep Neural Networks (DNN). While Logistic Regression was chosen for this study due to its mathematical interpretability, larger taxpayer datasets may benefit from the non-linear relationship mapping capabilities of advanced ensemble models.

2. Exploration of Real-Time API Data Stream Ingestion: This framework was designed for batch-processing historical transaction registries. Future studies should design real-time data ingestion pipelines that connect directly to banking transactional stream APIs, point-of-sale (POS) electronic fiscal devices, and customs import/export portals, allowing for real-time risk scoring and immediate compliance alerts.

3. Automated Algorithmic Drift and Retraining Monitoring: Research should be conducted on automating the model monitoring pipeline. Implementing drift-detection frameworks, such as the Population Stability Index (PSI) or Kullback-Leibler (KL) divergence, would allow the system to automatically trigger retraining pipelines when statistical data drift is detected, minimizing human administrative overhead.

4. Integration of Unsupervised Anomaly Detection: Because tax evasion tactics are highly adaptive, future research could explore hybrid models that combine supervised learning with unsupervised anomaly detection algorithms (such as Isolation Forests or Autoencoders). This approach would allow the system to flag completely novel, unseen evasion strategies that lack historical training labels.

5.5 Summary

This dissertation presented the end-to-end design, development, and empirical validation of a ten-stage supervised machine learning framework for risk-based audit selection within the South Sudan Revenue Authority.

Chapter One established the study's problem statement, focusing on the high error rates and subjective biases of legacy manual audit selection pipelines. Chapter Two reviewed the existing literature on machine learning architectures applied to public finance and tax governance. Chapter Three detailed the methodology, highlighting the selection of class-balanced Logistic Regression and the technical architecture of the Django web deployment. Chapter Four presented the empirical data analysis, demonstrating that the automated framework achieved a validation accuracy of 90.27% and a precision of 0.91, reducing audit selection errors to under 5.0% while mathematically correcting dual-currency scale imbalances.

Finally, Chapter Five interpreted these findings, drawing definitive conclusions on the effectiveness of AI-driven tax risk scoring. This chapter outlined action-oriented administrative and technical recommendations for immediate implementation within the SSRA and proposed theoretical pathways for future research in tax intelligence systems. Ultimately, this research provides a mathematically sound, practically validated blueprint for developing countries seeking to deploy data-driven solutions to curb revenue leakage, optimize resource allocation, and modernize public finance administration.

REFERENCES

Artemenko. (2017). From https://www.researchgate.net/publication/320969673_Tax_Risks_and_Their_Classification

Artemenko, D. (2017). *dmdocuments.* From https://ersj.eu/: https://ersj.eu/dmdocuments/2017-xx-3-b-41.pdf

ATAF. (2025). */news/2025-ataf-annual-meetings-outcome-statement/*. From https://ataftax.org: https://ataftax.org/news/2025-ataf-annual-meetings-outcome-statement/

Biber, & Edmund. (2010, April). */-/media/websites/imf/imported-full-text-pdf/external/pubs/ft/tnm/2010/_tnm1003.pdf.* From https://www.imf.org: https://www.imf.org/-/media/websites/imf/imported-full-text-pdf/external/pubs/ft/tnm/2010/_tnm1003.pdf

Bird, & Zolt. (2008). From https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1271350

Cents, F. (2025, August 12). *ai-accounting-terms/machine-learning-in-accounting/?referrer=www.google.com*. From financial-cents.com: https://financial-cents.com/ai-accounting-terms/machine-learning-in-accounting/?referrer=www.google.com

Chiloane, K., Isik, F., & Zita, E. (2024, November 20). *resources/isaca-journal/issues/2024/volume-6/digital-age-auditing-the-synergistic-effect-of-machine-learning-on-auditing-practices*. From https://www.isaca.org: https://www.isaca.org/resources/isaca-journal/issues/2024/volume-6/digital-age-auditing-the-synergistic-effect-of-machine-learning-on-auditing-practices

Choudhary, H. (2025, August 11). *Hybrid Approach to Tax Fraud Detection using Machine Learning.* From https://papers.ssrn.com/: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5366577#:~:text=The%20proposed%20hybrid%20model%2C%20integrating,allocation%20and%20streamline%20audit%20procedures.

Davis, W. (2024, September 27). */publications/tax-avoidance-tax-evasion-trade-misinvoicing-risks-senegal-mining-sector*. From https://resourcegovernance.org: https://resourcegovernance.org/publications/tax-avoidance-tax-evasion-trade-misinvoicing-risks-senegal-mining-sector

Dordevic, L., & Maftei, A. (2025, June 10). */imf-and-g20/2025/g20-background-note-enhancing-domestic-revenue-mobilization-through-strengthening-revenue-administra.pdf*. From publications/selected-issues-papers/issues/2025/06/06/domestic-revenue-mobilization-in-waemu-waemu-567482?: https://www.imf.org/en/publications/selected-issues-papers/issues/2025/06/06/domestic-revenue-mobilization-in-waemu-waemu-567482?

Emma, K. (2025). *posts/emmakellysafepaas_it-audit-the-limitations-of-manual-activity-7308788134945722368-wwny/*. From https://www.linkedin.com:

https://www.linkedin.com/posts/emmakellysafepaas_it-audit-the-limitations-of-manual-activity-7308788134945722368-wwny/

Glenn, A. (2023, November 24). *@glennlenormand/how-private-datasets-empower-ai-startups-in-the-modern-era-7bed2bb64e03*. From https://medium.com: https://medium.com/@glennlenormand/how-private-datasets-empower-ai-startups-in-the-modern-era-7bed2bb64e03

Hughes, L., & Riedel, S. (2025, May 7). *insights/facing-the-auditing-challenge-artificial-intelligence-machine-learning-and-robotics-in-financial-crime-compliance*. From https://ankura.com/: https://ankura.com/insights/facing-the-auditing-challenge-artificial-intelligence-machine-learning-and-robotics-in-financial-crime-compliance

Jain. (2024). From https://towardsdatascience.com/decision-trees-explained-6f8c6b5f6e9

Jain, A. (2024, September 12). From https://medium.com: https://medium.com/@abhishekjainindore24/all-about-decision-trees-80ea55e37fef

Jain, A. (2024, September 14). *@abhishekjainindore24/everything-about-random-forest-90c106d63989.* From https://medium.com/: https://medium.com/@abhishekjainindore24/everything-about-random-forest-90c106d63989

Jakkula, V. (n.d.). *cs5100f11/resources/jakkula.pdf.* From https://course.ccs.neu.edu: https://course.ccs.neu.edu/cs5100f11/resources/jakkula.pdf

Kagan, J. (2025, February 27). *Investopedia.com*. From https://www.investopedia.com: https://www.investopedia.com/terms/s/salestax.asp

Khwaja, Awasthi, & Loeprick. (2011). From https://openknowledge.worldbank.org/handle/10986/3405

Kinyanjui, M. (2025, April 30). *business/144154/kenya-lost-over-sh500-billion-to-tax-waivers-and-corruption-report*. From https://eastleighvoice.co.ke: https://eastleighvoice.co.ke/business/144154/kenya-lost-over-sh500-billion-to-tax-waivers-and-corruption-report

Kirchler. (2007). From https://www.routledge.com/The-Economic-Psychology-of-Tax-Behaviour/Kirchler/p/book/9780521683510

Mansour, M. B. (2024, November 19). */press/world-losing-half-a-trillion-to-tax-abuse-largely-due-to-8-countries-blocking-un-tax-reform-annual-report-finds/*. From https://taxjustice.net: https://taxjustice.net/press/world-losing-half-a-trillion-to-tax-abuse-largely-due-to-8-countries-blocking-un-tax-reform-annual-report-finds/

Martinez, A. L. (2025, June 08). *sol3/papers.cfm?abstract_id=5285760*. From https://papers.ssrn.com: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5285760

Mburu, P. (2024, April 25). */bd/economy/audit-reveals-kra-s-sh1-7trn-in-uncollected-taxes--4602272*. From https://www.businessdailyafrica.com: https://www.businessdailyafrica.com/bd/economy/audit-reveals-kra-s-sh1-7trn-in-uncollected-taxes--4602272

Mwangi, A. (2026, JANUARY 12). *bd/opinion-analysis/columnists/ai-in-cross-border-tax-administration-5324648*. From https://www.businessdailyafrica.com: https://www.businessdailyafrica.com/bd/opinion-analysis/columnists/ai-in-cross-border-tax-administration-5324648

OECD. (2010). From https://www.oecd.org/tax/forum-on-tax-administration/publications-and-products/compliance-risk-management-managing-and-improving-tax-compliance.htm

OECD. (2019). From https://www.oecd.org/tax/forum-on-tax-administration/publications-and-products/advanced-analytics-for-tax-administration.htm

OECD. (2025, September 15). */en/publications/2025/06/governing-with-artificial-intelligence_398fa287/full-report/ai-in-tax-administration_30724e43.html*. From https://www.oecd.org: https://www.oecd.org/en/publications/2025/06/governing-with-artificial-intelligence_398fa287/full-report/ai-in-tax-administration_30724e43.html

OECD. (2025). *content/dam/oecd/en/publications/reports/2025/06/governing-with-artificial-intelligence_398fa287/795de142-en.pdf*. From https://www.oecd.org: https://www.oecd.org/content/dam/oecd/en/publications/reports/2025/06/governing-with-artificial-intelligence_398fa287/795de142-en.pdf

OECD, & Bank, W. (2024;2025). From https://oecd.ai/en/gov/issues/tax-administration;https://www.worldbank.org/en/programs/govtech/brief/institutions-in-action-artificial-intelligence-for-the-selection-of-tax-audit-cases-in-georgia

Schnitt, D. (2023, October 27). */the-challenges-of-having-a-manual-accounting-system/*. From iqbackoffice: https://www.iqbackoffice.com/the-challenges-of-having-a-manual-accounting-system/

Schwikowski, M. (2025, April 9). *en/how-illegal-financial-flows-stifle-africas-growth/a-73868667*. From https://www.dw.com: https://www.dw.com/en/how-illegal-financial-flows-stifle-africas-growth/a-73868667

Shapiro, A. F. (1959). *495a4d/globalassets/.* From https://www.soa.org: https://www.soa.org/495a4d/globalassets/assets/files/static-pages/research/arch/2021/arch-2021-1-machine-learning.pdf

Soumaya, D. D. (2026). *Artificial Intelligence as a Tool for Tax Audit: A Legal–Economic Analysis of.*

Spinelli. (2025, November). */article/10.1007/s10797-025-09925-5*. From https://link.springer.com: https://link.springer.com/article/10.1007/s10797-025-09925-5

Standard Fusion. (2025, May 23). *blog/risk-based-approach-to-compliance-management*. From https://www.standardfusion.com: https://www.standardfusion.com/blog/risk-based-approach-to-compliance-management

The Independent. (2024, September 16). *ura-introduces-real-time-taxpayer-audit/*. From https://www.independent.co.ug: https://www.independent.co.ug/ura-introduces-real-time-taxpayer-audit/

Torgler, A. &. (2011). From https://www.sciencedirect.com/science/article/pii/S0047272710001591

Urama, P. K., & Ogunleye, D. E. (2023, October 5). */sites/default/files/2023/10/05/summary_for_policymakers_for_the_domestic_revenue_mobilization_in_africa_during_crisis_and_beyond_final.pdf*. From https://www.afdb.org: https://www.afdb.org/sites/default/files/2023/10/05/summary_for_policymakers_for_the_domestic_revenue_mobilization_in_africa_during_crisis_and_beyond_final.pdf

Vardan Baghdasaryan, H. D. (2022, Jan). *Improving Tax Audit Efficiency Using Machine.* From https://www.researchgate.net/: https://www.researchgate.net/publication/357662177_Improving_Tax_Audit_Efficiency_Using_Machine_Learning_The_Role_of_Taxpayer's_Network_Data_in_Fraud_Detection

Virani, A. (2025, 15 September). */post/risks-of-manual-accounting-and-legacy-systems*. From https://taxone.vyapar.com: https://taxone.vyapar.com/post/risks-of-manual-accounting-and-legacy-systems

White, J. (2025, December 12). *academy/lesson/regulatory-compliance-and-risk-management.html*. From https://study.com: https://study.com/academy/lesson/regulatory-compliance-and-risk-management.html

APPENDICES

3.0 Dataset

	PAYMENT DATE	REFERENCE NUMBER	INVOICE NUMBER	TYPE	TIN	NAME	CURRENCY	AMOUNT PAID	STATUS
2	31-12-2025	ETAXBQRHGGH366B8	LNKZVKD	tax_return	B20211102VDMGM4	UGANDA NATIONAL AIRLINES	USD	360.00	0
3	31-12-2025	ETAXD3HQN3PB4B4Q	EZEBXEN	tax_return	B20211102VDMGM4	UGANDA NATIONAL AIRLINES	USD	360.00	0
4	31-12-2025	ETAXRRPMLDJDBFQR	NGKMNKG	tax_return	B20211102VDMGM4	UGANDA NATIONAL AIRLINES	USD	360.00	0
5	31-12-2025	ETAXD82QJRBLN4FJ	LNKQQMV	tax_return	B20220804VWX8LX	GREEN SOM GENERAL TRADIN	SSP	1,161,000.00	0
6	31-12-2025	EXVXWN	KDGWZW	tax_return	B20220804VWX8LX	GREEN SON GENERAL TRADIN(	SSP	34,000.00	1
7	31-12-2025	ETAXQQ6DJG7BRRDG	JAPWQQA	tax_return	B20220804VWX8LX	GREEN SOM GENERAL TRADIN	SSP	170,000.00	0
8	31-12-2025	ETAXFNQPL2GPB2ND	QXRBBLK	tax_return	B20220804VWX8LX	GREEN SOM GENERAL TRADIN	SSP	1,563,966.00	0
9	31-12-2025	ETAXGF6N3HMPJMGD	YGLXXWM	tax_return	B20220804VWX8LX	GREEN SOM GENERAL TRADIN	SSP	1,593,900.00	0
10	31-12-2025	ETAXTN9MD6NRRJPQ	JAPWQQP	tax_return	B20220804VWX8LX	GREEN SOM GENERAL TRADIN	SSP	100,000.00	1
11	31-12-2025	GLBLNY	NDGXZV	tax_return	B20220804VWX8LX	GREEN SON GENERAL TRADIN(	SSP	34,000.00	1
12	31-12-2025	EQRQAM	KDGWPW	tax_return	B20220804VWX8LX	GREEN SON GENERAL TRADIN(	SSP	45,000.00	1
13	31-12-2025	ETAXD3226DDFHMDL	MXKZDAM	tax_return	B202307285J1YL0	FAST TECHNOLOGY LIMITED	SSP	66,654.00	1
14	31-12-2025	ETAXF78FJNDT66MN	BYLLVPE	tax_return	INGO202107195WQJZ5	HealthNet TPO South Sudan	USD	600.00	0
15	31-12-2025	ETAXPBRDGPNPR4FP	WYLLPGB	tax_return	INGO202107195WQJZ5	HealthNet TPO South Sudan	USD	120.00	0
16	31-12-2025	ETAXJTJ7PTM86MPJ	NGKKBDP	tax_return	INGO202107195WQJZ5	HealthNet TPO South Sudan	USD	5,802.00	0
17	31-12-2025	ETAXDM3DQPJPBH2G	BYLLVDM	tax_return	INGO202107195WQJZ5	HealthNet TPO South Sudan	USD	2,242.00	0
18	31-12-2025	ETAX97PTMTBN8P2Q	LNKKDAV	tax_return	B202411225J3X0Y	NOTOS LOUNGE CAFE BAR & R	SSP	3,622,819.00	0
19	31-12-2025	ETAX88D9N2PBQHH8	WYLLXZX	tax_return	B202411225J3X0Y	NOTOS LOUNGE CAFE BAR & R	SSP	1,012,000.00	0
20	31-12-2025	ETAXQMGHMH4BPBFQ	GXQYBZX	tax_return	B20220408VPJ3XK	AZ MEDIA AFRICA LTD	SSP	18,234.00	1

Figure 28: Showing dataset

2.0 Model Development

```
# Load dataset
df = pd.read_excel("gmt test dataset.xlsx")

# Convert payment dates
df["PAYMENT DATE"] = pd.to_datetime(df["PAYMENT DATE"], errors="coerce")

# Encode categorical currency values
df = pd.get_dummies(df, columns=["CURRENCY"], drop_first=False)

# Replace missing values
df = df.fillna(0)
```

Figure 29:Showing model development

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

model = LogisticRegression(
    class_weight="balanced",
    max_iter=2000
)

model.fit(X_train_scaled, y_train)
```

Figure 30:Showing model development

```
# Generate predictions
y_pred = model.predict(X_test_scaled)

# Model evaluation
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)

# Confusion matrix
cm = confusion_matrix(y_test, y_pred)

print(classification_report(y_test, y_pred))
print(cm)
```

Figure 31:Showing model development

3.0 Approval from SSRA



Republic of South Sudan
SOUTH SUDAN REVENUE AUTHORITY
Commissioner for Information Technology & Digital Transformation

Date: April 20th, 2026

Ref: SSRA/C/ITDT/2026

The Dean
Faculty of Science and Technology
Victoria University
Kampala, Uganda

RE: AUTHORIZATION TO CONDUCT RESEARCH AND DATA COLLECTION

Dear Sir,

Reference is made to your letter dated **10 April 2026** requesting authorization for **Mr. Anyieth Aguek Ayuen** to conduct data collection at the South Sudan Revenue Authority (SSRA) in support of his Master's research titled *"A Supervised Machine Learning Framework for Risk-Based Audit Selection in the South Sudan Revenue Authority (SSRA)."*

We are pleased to inform you that **approval** is hereby **granted** for the proposed data collection, subject to compliance with SSRA's confidentiality, data protection and information security policies. The researcher is requested to coordinate with the Information Technology & Digital Transformation Division to facilitate access to the required information and schedule engagements with the relevant departments.

Please note that any information obtained during the study shall be used strictly for academic purposes and shall not be disclosed to unauthorized parties without the prior written consent of SSRA.

We wish Mr. Anyieth success in his research and look forward to the findings, which may contribute to strengthening data-driven tax audit and revenue administration within SSRA.

Yours faithfully,

Garang Maker Thiong
Commissioner,
Information Technology & Digital Transformation
South Sudan Revenue Authority

South Sudan Revenue Authority Headquaters, Jebel Area, Juba, South Sudan
Email: garang.maker@nra.gov.ss **web:** www.nra.gov.ss

Figure 32: Showing authorization from SSRA